

# When Does Automating AI Research Produce Explosive Growth? Feedback Loops in Innovation Networks

Tom Davidson\* Basil Halperin<sup>†</sup> Thomas Houlden<sup>‡</sup> Anton Korinek<sup>§</sup>

October 2026

## Abstract

AI labs increasingly use AI itself to accelerate AI research, creating a feedback loop that could produce an intelligence explosion. We develop a general semi-endogenous growth model with an innovation network and AI-driven automation. Technological feedback loops (spillovers across sectors) and economic feedback loops (higher output financing further investment) offset ideas getting harder to find. We derive a clean analytical condition on the level of automation – and, with bottlenecks, its speed – required to achieve hyperbolic growth. In a simple simulation calibrated to AI progress, fully automating software research plus 5% automation elsewhere generates a singularity within six years.

**Keywords:** AI, economic growth, automation, innovation networks, feedback loops, intelligence explosion, singularity

---

\*Forethought: tom@forethought.org

<sup>†</sup>University of Virginia: basilh@virginia.edu

<sup>‡</sup>Columbia University: twh2125@columbia.edu

<sup>§</sup>University of Virginia, EconTAI and Anthropic Institute: econtai@virginia.edu

We thank Dan Carey, Tom Cunningham, Chad Jones, Ezra Oberfield (discussant), Philip Trammell, Parker Whitfill, and Cheryl Wu, as well as participants at conferences and seminars hosted by Oxford, the Hong Kong Monetary Authority, Chicago Booth, the International Monetary Fund, the BSE Summer Forum, IESE, and the NBER Summer Institute for helpful comments and discussion.

# 1 Introduction

“Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an ‘intelligence explosion’, and the intelligence of man would be left far behind.”

— [Good \(1966\)](#), Speculations Concerning the First Ultraintelligent Machine

AI has the potential to automate many different kinds of work: customer support, coding, marketing, and many other tasks. However, a widespread belief among leading AI researchers is that automation of AI research itself in particular will have a transformative economic impact. Central to this thesis is the argument that such *recursive self-improvement* – where AI systems become increasingly capable of designing and improving themselves – creates a feedback loop leading to an “intelligence explosion” and rapid economic growth ([Yudkowsky, 2013](#)). OpenAI bluntly declares its goal of developing such technology “by March of 2028” ([OpenAI, 2025](#)).

Whether such recursive self-improvement actually produces explosive economic growth depends on two economic forces:

1. **Diminishing returns.** A self-improving process can generate hyperbolic growth (“explode”), but – contra [Good \(1966\)](#) – it need not do so. Whether or not a recursively self-improving process explodes depends critically on the strength of *diminishing returns*. Formally, a process with a positive feedback loop  $\frac{dy}{dt} = y^{1-\beta}$  will not explode if there are diminishing returns, i.e. if  $\beta > 0$ . Intuitively, a self-improving AI may “pick all the low-hanging fruit first” and find it increasingly difficult to make algorithmic progress, and as a result progress may be subexponential or even stagnate. In the economics literature, this corresponds to models where “ideas get harder to find” ([Bloom et al., 2020](#)).
2. **Bottlenecks.** Even if one process in the economy achieves explosive growth, this does not necessitate that aggregate growth explodes: progress in one sector may be bottlenecked by slow progress in other sectors ([Aghion, Jones, and Jones, 2019](#); [Jones, 2025](#)). Intuitively, even if AI were capable of producing infinite left shoes, “total shoe output” would still be bottlenecked by production of right shoes. Formally, this is captured by *complementarity*: in its most extreme form, total output is the minimum of two components,  $y = \min\{y_1, y_2\}$ .

**This paper.** We develop a framework for analyzing how AI-driven automation interacts with both forces, and identify the conditions under which feedback loops generated by automation tip the economy into explosive growth. We begin with a benchmark model without bottlenecks to isolate whether automation can overturn diminishing returns in research; we then show how fast automation must progress for this result to survive when tasks are gross complements.

Our framework is a general semi-endogenous growth model with multiple technologies, where each technology can generate spillovers to others through a network. This structure may be of independent interest to the growth literature but is particularly relevant for our analysis of the AI R&D process. We allow for automation in the ideas production function ([Aghion, Jones, and Jones, 2019](#)) and examine how automation in one research sector spills through the innovation network and interacts with economic growth.

The model identifies two distinct channels through which automation generates explosive dynamics, and these channels mutually reinforce each other. The first is *technological* feedback loops across the innovation network. In the canonical semi-endogenous model, there is one research sector; our model instead features a network of heterogeneous research sectors, where innovations in one sector spill over to increase the rate of innovation in other sectors ([Liu and Ma, 2024](#); [Chen, Liu, and Ma, 2026](#)). Such spillovers are important for capturing, for example, the feedback loop between better software and better hardware: better computer chips allow for the design of better AI models, while better AI models in turn are used to help to design yet better computer chips ([Mirhoseini et al., 2021](#)), and so on. The second channel is an *economic* feedback loop, in which higher output finances investment that drives further economic growth. The classic example is capital accumulation: higher output leads to more investment, which produces yet more output. In our setting, this channel is particularly important: AI hyperscaler investment is over 2% of US GDP per year and has been growing 70% per year ([Epoch AI, 2026](#)). A larger economy supports more investment.

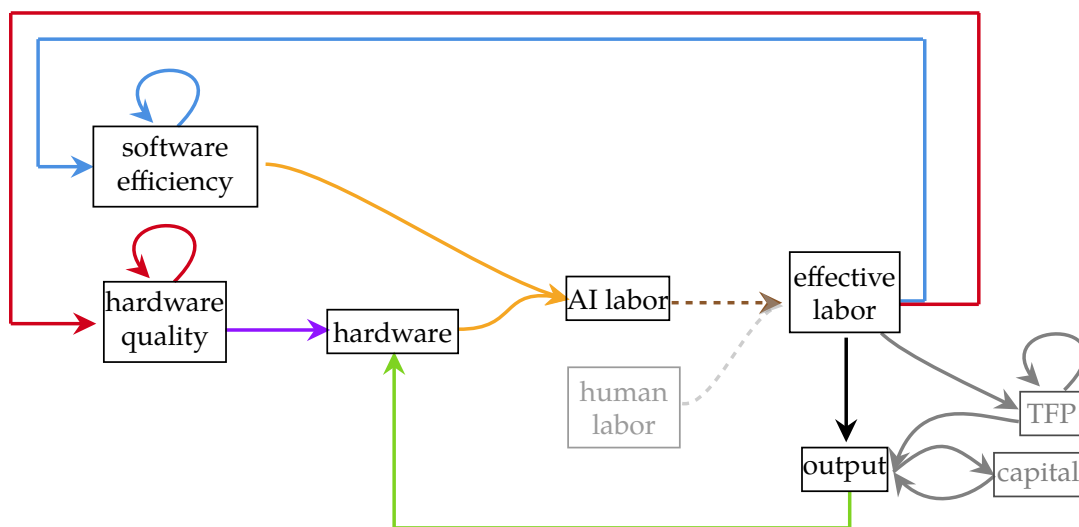
**Analytical insights.** The general model produces a simple analytical condition under which technological and economic feedback loops give rise to balanced or explosive growth, along with several broad, interpretable insights. First, technological feedback loops – spillovers across research sectors – can offset diminishing returns in the production of ideas. They reduce the degree to which ideas get harder to find, so that if such spillovers are improperly ignored, a system may be estimated to be non-explosive when in reality spillovers from other sectors tip it into explosive dynamics.

Second, automation creates economic feedback loops, which similarly counteract di-

minishing returns. Automation means that a task previously performed by human labor is instead performed by machines, i.e. capital. Human labor does not generate an economic feedback loop in the modern era: higher GDP does not result in a higher population. Machines, on the other hand, do: higher GDP leads to the construction of more machines. Replacing human labor with capital therefore counteracts the diminishing returns in the production of ideas by creating a new feedback loop where one did not exist previously.

Third, technological and economic feedback loops amplify each other. AI research results in higher output, which helps fund further investment in AI research, which in turn drives further technological progress.

**Application: AI automation with software & hardware feedback loops.** We apply the general framework to study an economy with AI-driven automation, modeled to match key features of modern AI development, to analyze the titular question of how automation of AI research affects economic growth. While the model is rich and complex, it can be studied analytically and delivers closed-form results. Figure 1 provides an overview of the structure.



**Figure 1:** Model of AI automation with software & hardware feedback loops. Dashed lines indicate potential bottlenecks.

Our starting assumption is that AI can automate a fraction of tasks across different sectors in the economy. AI itself is a combination of a nonrivalrous idea, “software” (AI algorithms), and a kind of capital, “hardware” (computer chips like Nvidia GPUs). Software progress follows a canonical ideas production function. Computer chip hardware, meanwhile, accumulates like any form of capital, augmented by investment-specific technical change: “hardware quality” is another standard nonrivalrous input following an

ideas production function. In addition to software and hardware quality, we have a third innovation sector: a “general” research sector creating new ideas for goods production, as in standard models. Production of new ideas in any of the three innovation sectors may be performed by some combination of human labor and AI. Finally, goods output likewise may be produced by a combination of humans and AI.

We then use the model to illustrate how automation of both research and production can amplify existing feedback loops or spawn them where they did not exist before. In particular, the model predicts hyperbolic growth will arise under the simple condition:<sup>1</sup>

$$f_Y + f_S \left( \frac{1}{\beta_S} \right) + f_H \left( \frac{1}{\beta_H} \right) + f_A \left( \frac{1}{1 - \alpha} \right) \left( \frac{1}{\beta_A} \right) > 1 \quad (1)$$

Here, the subscripts denote sectors: final output  $Y$ , software research  $S$ , hardware research  $H$ , and general research  $A$ . The term  $f_i \in [0, 1]$  is the fraction of tasks in each sector that have been automated by AI. The term  $\beta_i > 0$  is the degree of diminishing returns in each research sector.  $1 - \alpha$  is the pre-automation labor share in production of output.

The explosion condition (1) implies that growth is hyperbolic if the sum of the strengths of feedback loops exceeds unity. In particular, the relevant feedback loops are: (i) the pure economic feedback loop (which will be strengthened with  $f_Y$ , the level of final output automation); (ii) the feedback loops induced through technological and economic channels as automation accelerates research (the next three terms,  $f_i$  divided by diminishing returns  $\beta_i$ , for each research sector).<sup>2</sup>

This condition by itself does not pin down an exact growth path or the timing of a growth explosion. Rather, it specifies the condition that determines whether growth will *eventually* explode. Hence, we think of this condition as a line in the sand: supposing other parameters are fixed, how much automation  $f_i$  must be achieved to tip the economy into an explosive growth regime?

**Empirical insights & quantitative results.** The explosion condition highlights the critical importance of measuring diminishing returns sector by sector. Bloom et al. (2020) estimate that in the economy as a whole, ideas become sharply harder to find, with  $\beta_A = 3.1$ . The hardware sector exhibits the same phenomenon – more and more researchers are required to maintain the pace of Moore’s Law – but the magnitude is dramatically smaller:  $\beta_H = 0.2$ . Indeed, hardware shows the smallest degree of diminishing returns of any sector studied. Estimating diminishing returns in software is more challenging; we adopt a

<sup>1</sup>For simplicity, the expression here assumes no “parallelization penalty”; (59) generalizes.

<sup>2</sup>In the case of general innovation,  $A$ , diminishing returns are modulated by the labor share,  $1 - \alpha$ .

relatively conservative calibration of  $\beta_S = 1$ , drawing on [Erdil, Besiroglu, and Ho \(2024\)](#)’s preferred estimate from chess engine data, while noting that other recent estimates point to lower values ([Ho and Whitfill, 2025](#)).

Applying this calibration to the explosion condition produces three analytic findings that only depend on the estimates of diminishing returns and the economy-wide labor share. First, the threshold is reachable: 13% automation across all sectors is sufficient to push the economy into the explosive regime, and 17% suffices when only software and hardware research are automated. Second, automation of hardware research has particularly large effects, because diminishing returns to research in hardware are roughly five times lower than in software and fifteen times lower than in general research. Automating one task in chip design moves the economy as much as ten tasks in general research; 20% automation of hardware alone is enough to cross the threshold. Third, automating software in isolation is approximately at the knife-edge: fully automating software research without automating any other part of the economy just reaches the explosive growth threshold. A small push elsewhere is sufficient to tip the system.

We also simulate how quickly the singularity arrives once the threshold is crossed, an exercise which requires additional calibration to current algorithmic and hardware progress ([Epoch AI, 2026](#)). In our baseline simulation – a stylized exercise rather than a forecast – an ‘automation shock’ involving full automation of software R&D and just 5% automation across the rest of the economy causes a singularity in six years.

**The role of bottlenecks.** In our final section, we generalize our framework beyond the unit elasticity of substitution case to allow for bottlenecks and show how *continually progressing* automation – dynamically evolving  $f$  terms – can overcome the constraining effects of such gross complementarity. We show that there is a race between the constraining effects of bottlenecks versus the amplifying effects of automation and analytically derive a sufficient condition on the speed of automation that restores the explosion dynamics from our baseline model.

**Related literature.** Several studies have investigated the possibility of AI and automation leading to transformative growth. [Aghion, Jones, and Jones \(2019\)](#) characterize the conditions under which a single-sector system capable of recursive self-improvement exhibits hyperbolic growth. They also embed such self-improving technologies in a macroeconomic setting where capital accumulation and technological progress reinforce each other. [Jones \(2025\)](#) quantitatively studies the role of bottlenecks in preventing explosive growth, and [Jones and Tonetti \(2025\)](#) estimate the size of these bottlenecks historically and simulate their future impact.

We advance this line of work in three main directions. First, we generalize the analysis to systems whose progress depends on a network of multiple technologies with heterogeneous degrees of diminishing returns. Second, we integrate technological heterogeneity with economic feedback loops, showing how output-technology complementarities amplify the possibility of hyperbolic growth. Third, we use this richer framework to derive a threshold for automation that marks the transition from balanced to hyperbolic growth and study it quantitatively.

[Erdil et al. \(2025\)](#) model the growth implications of AI in a computational integrated assessment model. Our framework is simpler and sufficiently tractable to capture similar core dynamics analytically. This allows us to explicitly characterize feedback loops and their interactions and to derive an explicit explosion threshold.

Networked models of technological progress have also been applied to understand optimal R&D allocations ([Liu and Ma, 2024](#)), heterogeneity in sectoral productivity growth ([Ngai and Samaniego, 2011](#)), and the dynamics of clean innovation ([Donald, 2023](#)). In comparison to these papers, we embed automation into the innovation network, allow for the possibility of explosive growth, and apply the framework to study AI-driven automation of software and hardware research.<sup>3</sup>

**Outline.** Section 2 develops a set of simple models illustrating the core economic forces at work. Sections 3 and 4 develop the general theory – the fully general network model and AI-driven automation – and contain the most technically demanding material in the main text. Readers primarily interested in the economic intuition and quantitative results may skip these sections on a first reading and proceed directly to section 5, which presents our applied model, and section 6, which extends the results to a world with bottlenecks. Section 7 concludes.

## 2 A sequence of simple models

As is typical in networked settings, our general model is fairly complicated. This section presents a sequence of simple models to highlight the core economic forces at work in the general model. We draw out four lessons:

1. Diminishing returns prevent growth explosions.
2. Innovation networks (technological feedback loops) introduce spillovers and offset diminishing returns.
3. Economic feedback loops also introduce spillovers and offset diminishing returns.

---

<sup>3</sup>[Cunningham et al. \(2026\)](#) build on our applied model and work out several extensions.

4. AI-driven automation introduces new feedback loops (or amplifies existing ones).

## 2.1 Lesson one: Diminishing returns prevent growth explosions

The simplest possible formalism for I.J. Good’s concept of an intelligence explosion is  $\dot{S}_t = S_t$ , where  $S$  is the level of intelligence or the level of “software productivity”, and dots indicate time derivatives. This equation says that when the *level* of intelligence  $S_t$  is low, the *rate of change* of intelligence  $\dot{S}_t$  is also low; and when the level of intelligence is high, the rate of change of intelligence is also high.

We can generalize this process to:

$$\dot{S}_t = S_t^{1-\beta} \quad (2)$$

With this generalization, we can observe that – contra [Good \(1966\)](#) and many since – there need not be an “explosion” from a recursively self-improving process. In particular, if  $\beta < 0$ , so that there are increasing returns, then there is a literal mathematical singularity:  $S_t$  approaches infinity in finite time. On the other hand, if  $\beta = 0$ , the process exhibits exponential growth, and if  $\beta > 0$ , so that there are diminishing returns, the process is subexponential or even sublinear. This is a simple reminder of the importance of diminishing returns in preventing runaway feedback processes.

Equation (2) is in fact the canonical form of the ideas production function for modeling the growth of productivity (abstracting from research inputs for now), and it can be easily embedded in an economic growth model to think about the relationship between intelligence explosions and economic explosions. The simplest possible case has a goods production function as follows, assuming a constant labor supply  $L$ :

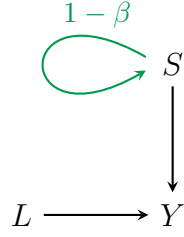
$$Y_t = S_t L^{1-\alpha} \quad (3)$$

Output is produced with labor input (subject to potentially diminishing returns,  $\alpha \in [0, 1]$ ) augmented by software capabilities.

The simple economy of (2)-(3) clearly features an *economic* explosion – infinite output  $Y$  in finite time – if and only if there is an intelligence explosion, i.e.,

$$\beta < 0 \quad (4)$$

This model is summarized in figure 2.



**Figure 2:** Recursive self-improvement explodes if and only if there are increasing returns ( $\beta < 0$ ).

## 2.2 Lesson two: Innovation networks introduce spillovers and offset diminishing returns

It is well-known that higher quality computer chips are used by AI researchers to write better algorithms; additionally, it is increasingly the case that those better AI algorithms are used in turn to help design better chips. “AlphaChip” (Mirhoseini et al., 2021) from Google DeepMind is a striking example of this phenomenon. AlphaChip is a reinforcement learning method for designing chip layouts, and it is reported to have been used in designing every new generation of Google’s Tensor Processing Unit chip since 2020 and to be responsible for a growing share of the ‘floorplan’ for each generation of chip (Goldie and Mirhoseini, 2024).

To capture this, we modify the baseline semi-endogenous growth model of (2)-(3) to introduce a two-sector, *networked* semi-endogenous growth model. Continuing to denote  $S$  as software productivity, denoting  $H$  as hardware quality, and dropping time subscripts to ease notation,<sup>4</sup>

$$\dot{S} = S^{1-\beta_S} H^{\phi_S} \quad (5)$$

$$\dot{H} = H^{1-\beta_H} S^{\phi_H} \quad (6)$$

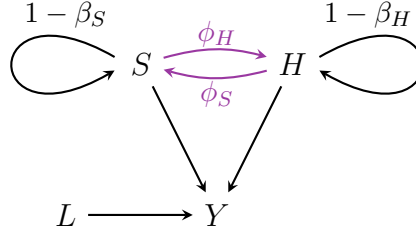
$$Y = (SH)^{1/2} L^{1-\alpha} \quad (7)$$

Here,  $\beta_S$  now denotes diminishing returns within software (“ideas getting harder to find”); likewise,  $\beta_H$  the same within hardware. Meanwhile  $\phi_S \geq 0$  reflects the *technological spillovers* from hardware quality to software improvements, and vice versa for  $\phi_H$ . This nests the dynamics of software in the prior model under  $\phi_S = 0$ .

In this networked example, we now have three feedback loops, which can be seen by

---

<sup>4</sup>The choice of equal-weighted Cobb-Douglas aggregation of  $S$  and  $H$  in the goods production function (7) is not essential.



**Figure 3:** Innovation networks introduce cross-sector spillovers (here, from  $S \rightarrow H$  and  $H \rightarrow S$ ), offsetting diminishing returns.

physically tracing all possible simple “loops” in figure 3:

1. Recursive self-improvement within software, as before, governed by  $\beta_S$ .
2. Recursive self-improvement within hardware quality, governed by  $\beta_H$ .
3. Spillovers across the innovation network, intermediated via  $\phi_H$  and  $\phi_S$ .

The spillovers across the innovation network are summarized by the interaction matrix: the matrix collecting the exponents in (5)-(6):

$$\begin{bmatrix} 1 - \beta_S & \phi_S \\ \phi_H & 1 - \beta_H \end{bmatrix}$$

The system can now explode in two ways. First, analogously to the prior single-sector example, the system explodes if either recursive self-improvement loop is strong enough on its own,  $\beta_S < 0$  or  $\beta_H < 0$ .

Second, the system explodes via the spillover loop *if the interaction between the two loops is strong enough*. It turns out that, mathematically, this occurs if the interaction matrix has an eigenvalue greater than one. This generalizes the single-sector condition that the exponent in the law of motion,  $1 - \beta$ , is greater than one. In turn, if returns are diminishing, it can be shown that the eigenvalue condition holds here if and only if:

$$\underbrace{\beta_S \cdot \beta_H}_{\text{diminishing returns}} < \underbrace{\phi_S \cdot \phi_H}_{\text{spillovers}} \quad (8)$$

Condition (8) implies that spillovers effectively offset diminishing returns. For example, suppose  $\beta_H = 1$ , so that the only difference from the model in section 2.1 relevant for explosive dynamics is spillovers. Then the condition (8) for explosive growth becomes

simply  $\beta_S < \phi_S \phi_H$ . Thus, explosive growth no longer requires increasing returns,  $\beta_S < 0$ , but now can occur if diminishing returns are mild,  $\beta_S \in [0, \phi_S \phi_H)$ .

### 2.3 Lesson three: Economic feedback loops introduce spillovers and offset diminishing returns

In our full AI economy model, we combine the technological feedback loops from an innovation network, as in section 2.2, with *economic* feedback loops. An “economic” feedback loop refers to a feedback loop in which higher output is involved.

The most basic economic feedback loop is a Solow model without technological progress: normalizing the population to one, output is produced as  $Y = K^\alpha$  and capital accumulates as  $\dot{K} = aY - \delta K$ , where  $a$  is a constant savings rate and  $\delta$  the depreciation rate. This model is well-known to feature explosive growth if there are increasing returns to capital in production –  $\alpha > 1$  – or in the language used here, if the economic feedback loop is sufficiently strong.<sup>5</sup>

In the rest of this subsection, we illustrate an economic feedback loop using the canonical single-sector semi-endogenous growth model, with capital instead of labor in the ideas production function:

$$\dot{A} = A^{1-\beta_A} \cdot (\kappa_A K)^\gamma \quad (9)$$

$$Y = A \cdot L^{1-\alpha} \cdot (\kappa_Y K)^\alpha \quad (10)$$

$$\dot{K} = aY - \delta K \quad (11)$$

Here,  $A$  is a general productivity term produced from general research in the economy, replacing the previous  $S$  term. ( $\kappa_A$  is the share of capital used for research;  $\kappa_Y \equiv 1 - \kappa_A$  is the share used in production.) This model is summarized in figure 4.

The same logic can be applied as before, with the same condition on the eigenvalues of the interaction matrix. The condition implies the system explodes if:

$$\underbrace{\beta_A \cdot (1 - \alpha)}_{\text{diminishing returns}} < \underbrace{\gamma \cdot 1}_{\text{spillovers}} \quad (12)$$

This condition is exactly analogous to the condition of lesson two, (8), and highlights that when higher productivity  $A$  increases output, then if this output in turn can be invested to produce yet further research advances ( $\gamma > 0$ ), then explosive growth is more likely.

<sup>5</sup>And if initial capital is sufficiently plentiful,  $K_0 > (\delta/a)^{1/(\alpha-1)}$ .

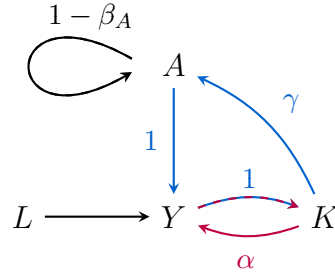


Figure 4: Economic feedback loops (here, both  $A \rightarrow Y \rightarrow K \rightarrow A$  and  $K \rightarrow Y \rightarrow K$ ) finance further investment (i.e. expand  $K$ ), offsetting diminishing returns.

Notably, a standard calibration of (12) would imply a lack of explosive growth. Using  $\beta_A = 3.1$  as the extent to which ideas are getting harder to find in the economy as a whole (Bloom et al., 2020),  $1 - \alpha = 0.6$  as the labor share in production, and  $\gamma = 0.1$  as the capital share in R&D for the economy as a whole (Besiroglu, Emery-Xu, and Thompson, 2024), we find that the explosion condition is far from being met. However, as the next subsection shows, automation could change this.

## 2.4 Lesson four: AI-driven automation introduces new feedback loops

Finally, we come to the role of automation, which is critical to our titular question. Here, we discuss AI-driven automation of ideas production, which triggers new *technological* feedback loops, and hold the quantity of capital constant; in section 4, we also allow for AI to automate the production of final goods and for capital to accumulate, which triggers new *economic* feedback loops as well.

In traditional models of automation of innovation (Aghion, Jones, and Jones, 2019), capital replaces labor in the effective ideas production function. Formally: again letting  $S$  denote the level of software efficiency,<sup>6</sup>

$$\dot{S} = S^{1-\beta} \cdot \hat{L} \quad (13)$$

$$\hat{L} \equiv L^{1-f} \cdot K^f \cdot \xi \quad (14)$$

Here,  $\hat{L}$  is *effective labor*: the effective amount of software innovation done by humans and machines. Importantly,  $f \in [0, 1]$  measures the share of tasks that have been automated

<sup>6</sup>Underlying this, effective labor is a task aggregate,  $\hat{L} = T \prod_{i=1}^T X_i^{1/T}$ , for  $T$  different tasks  $X_i$  bundled together with equal weights with an elasticity of substitution equal to one. Tasks  $i = 1, \dots, I$  are performed by capital, with the remainder performed by labor. Given aggregate labor  $L$  and sufficiently plentiful aggregate capital  $K$ , inputs are optimally spread equally across tasks. Defining  $f \equiv I/T$  results in equation (14).

(while  $\xi$  is an unimportant constant). For studying explosive dynamics, automation can be understood as simply increasing  $f$ , shifting production weight from  $L$  to  $K$ .

In this traditional model, automation does not affect whether there will be an explosion. An innovation explosion still requires increasing returns,  $\beta < 0$ , regardless of the level of automation.

We instead introduce the notion of *AI-driven automation*, in which technology-augmented capital replaces labor.<sup>7</sup> We denote technology-augmented capital as “AI labor”. The model then becomes:

$$\dot{S} = S^{1-\beta} \cdot \hat{L} \quad (15)$$

$$\hat{L} \equiv L^{1-f} \cdot (L_{\text{AI}})^f \cdot \xi \quad (16)$$

$$L_{\text{AI}} \equiv S \cdot K \quad (17)$$

The interpretation is that AI performs work by using a given quantity of hardware capital  $K$  (“compute”); and the amount of work a given quantity of compute can perform depends on the level of software efficiency  $S$ . That is, fixing a particular automated task within software research, AI in 2026 is more productive than AI in 2020 for a given quantity of compute.<sup>8</sup>

AI-driven automation is qualitatively different from traditional automation in that it introduces a new feedback loop, and thus makes explosive growth more likely. The new technological feedback loop arises because software – as embedded in AI – is used to produce software improvements. This ability of AI-driven automation to trigger a new feedback loop, or to strengthen an existing one, is visualized in figure 5.

As a result, instead of requiring increasing returns ( $\beta < 0$ ), there is explosive growth in software efficiency when:

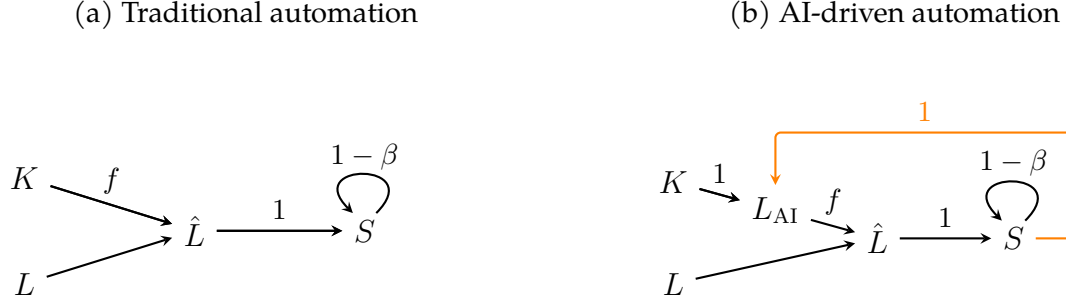
$$\beta < f \quad (18)$$

This is derived from simple substitution. Thus, higher automation  $f$  makes explosive growth more likely. Notice that if the level of diminishing returns  $\beta$  is less than 1, a sufficient rise in automation  $f$  (which is by definition bounded above by 1) could tip the system into explosive dynamics.

---

<sup>7</sup>Outside the economics literature, [Davidson \(2023\)](#) adopts this notion of automation, without contrasting it to the standard notion. Within the economics literature, this is a particular endogenous form of what [Acemoglu and Restrepo \(2019\)](#) call “deepening of automation”.

<sup>8</sup>Note that this is distinct from investment-specific technical change, which would be a notion of “hardware quality” as we introduce in section 5. Investment-specific technical change improves the flow of new investment into hardware; improved software affects the entire stock of hardware.



**Figure 5:** AI-driven automation creates the technological feedback loop  $S \rightarrow L_{AI} \rightarrow \hat{L} \rightarrow S$ , as the degree of automation rises from zero to  $f > 0$  (or strengthens it, if it already exists). Traditional automation does not create (or strengthen) any feedback loops.

## 2.5 Summary

Using this last condition (18), we can now return to the boxed equation shown in the introduction for our applied model of section 5, equation (1), to provide some intuition for its origin. Recall that condition:

$$f_Y + f_S \left( \frac{1}{\beta_S} \right) + f_H \left( \frac{1}{\beta_H} \right) + f_A \left( \frac{1}{1 - \alpha} \right) \left( \frac{1}{\beta_A} \right) > 1$$

Now consider the condition we derived in lesson four, (18). In the empirically realistic case of diminishing returns, (18) can be rearranged as:

$$f \cdot \left( \frac{1}{\beta} \right) > 1 \tag{19}$$

The level of automation, scaled down by the degree of diminishing returns, must be greater than one. The boxed condition can be seen to contain essentially the same condition, but with a left-hand side that is summed across four sectors. This is because condition (19) was derived in a one-sector economy while our applied model will feature AI-driven automation of four different sectors (goods production  $Y$ , software  $S$ , hardware quality  $H$ , and “general ideas”  $A$ ).<sup>9</sup>

**The rest of the paper.** The sequence of simple models above illustrates the core economic forces at work. They also show a striking degree of formal mathematical parallelism. We

<sup>9</sup>General ideas  $A$  is modulated by the factor  $1/(1 - \alpha)$  because it is Hicks-neutral rather than labor-augmenting. That is,  $S$  and  $H$  enter output as effective labor, and output has an elasticity of  $1 - \alpha$  with respect to labor; meanwhile, output has an elasticity of 1 with respect to  $A$ .

now turn to a general framework that explains the deeper underlying structure.

### 3 A general framework for hyperbolic growth

In this section, we present a general framework to think about the ideas introduced in section 2. We begin in section 3.1 by introducing a general innovation network, with spillovers generating arbitrary possible feedback loops between technologies, to consider the necessary and sufficient conditions for hyperbolic growth from technological feedback loops alone. We then embed this model of networked technological progress into an economic environment in section 3.2, demonstrating how economic feedback loops can be isolated analytically in their contribution to balanced or explosive growth. Section 4 introduces AI-driven automation.

Readers interested in applied results may wish to skip to section 5, where the general framework is applied to a model of the AI research production network.

#### 3.1 Technological feedback loops

Consider an economy with  $N$  different technological sectors, indexed  $i \in I \equiv \{1, \dots, N\}$ . We model technological progress as an innovation network, with sectoral ideas production functions

$$\dot{A}_i = v_i R_i^{\lambda_i} A_i^{1-\beta_i} \prod_{j \neq i} A_j^{\phi_{i,j}} \quad (20)$$

in which progress in each sector benefits from spillovers from other sectors. Here,  $A_i$  is the level of technology in sector  $i$ ,  $R_i$  is the quantity of research inputs (e.g. labor or capital), and  $v_i > 0$  is a constant parameter. We suppress time subscripts; technologies  $A_i$  and research inputs  $R_i$  are time-dependent. Assume for now that research inputs  $R_i$  grow exogenously, at some constant rate  $g_{R_i} > 0$ . There are intratemporal diminishing returns to parallel research input,  $\lambda_i \in (0, 1]$ . The parameter  $\beta_i$  captures the degree of intertemporal diminishing returns to research within a given sector as introduced in Jones (1995). We impose  $\beta_i > 0$ , so “ideas are getting harder to find”, which rules out hyperbolic growth arising trivially from increasing returns within any individual technology. There are non-negative spillovers across innovation sectors,  $\phi_{i,j} \geq 0$ , for  $i \neq j$ . If spillovers are uniformly zero,  $\phi_{i,j} = 0$ , then this specification recovers the canonical semi-endogenous ideas production function.

**Balanced growth and the technological feedback matrix.** To build intuition for the role of technological spillovers and feedback loops, consider first the case where a balanced growth path exists, on which each technology grows at a constant rate  $g_{A_i}^{\text{BGP}}$ . Log-differentiating the ideas production function (20) along such a path gives

$$g_{A_i}^{\text{BGP}} = \frac{\lambda_i}{\beta_i} g_{R_i} + \sum_{j \neq i} \frac{\phi_{i,j}}{\beta_i} g_{A_j}^{\text{BGP}} \quad (21)$$

Thus, technological growth in a sector is driven both by growth in its own research inputs ( $g_{R_i}$ ) and by technological progress elsewhere in the innovation network (the  $g_{A_j}$  terms).

It will be useful to define labels for two of the terms of (21), as they will appear repeatedly in our analysis. First, define:

$$r_i \equiv \frac{\lambda_i}{\beta_i} \quad (22)$$

From (21), it can be seen that the term  $r_i$  is the direct elasticity of the level of technology  $A_i$  with respect to the level of research inputs  $R_i$  on the BGP.<sup>10</sup> For our purposes, we can typically think of this  $r_i$  term as simply being inversely related to diminishing returns  $\beta_i$ . Collect these terms into the matrix  $\mathbf{r} = \text{diag}(r_1, \dots, r_N)$ .

Second, define the technological feedback matrix  $\mathbf{F}^A \in \mathbb{R}_{\geq 0}^{N \times N}$  by

$$F_{i,j}^A \equiv \begin{cases} \frac{\phi_{i,j}}{\beta_i}, & \text{if } i \neq j, \\ 0, & \text{if } i = j. \end{cases} \quad (23)$$

An entry  $F_{i,j}^A$  measures the strength of the technological spillovers  $\phi_{i,j}$ , relative to the degree of diminishing returns  $\beta_i$ . This feedback matrix is a critically important object.

Stacking sectoral technological growth rates into a vector  $\mathbf{g}_A^{\text{BGP}}$  and research input growth rates into  $\mathbf{g}_R$ , then the BGP (21) can be written as  $\mathbf{g}_A^{\text{BGP}} = \mathbf{r} \mathbf{g}_R + \mathbf{F}^A \mathbf{g}_A^{\text{BGP}}$ . This shows how technological spillovers  $\mathbf{F}^A$  amplify growth along a balanced growth path. Remark-

---

<sup>10</sup>That is, in the case without spillovers, on the BGP  $\frac{d \log A_i}{d \log R_i} = \frac{d \log A_i / dt}{d \log R_i / dt} = \frac{g_{A_i}}{g_{R_i}} = r_i$ . Alternatively, write the BGP paths as  $A_{it} = A_{i0} e^{g_{A_i} t}$  and  $R_{it} = R_{i0} e^{g_{R_i} t}$ . Then, eliminating  $t$ ,  $A_{it} = A_{i0} (R_{it}/R_{i0})^{g_{A_i}/g_{R_i}}$ . Thus, the elasticity of  $A_{it}$  with respect to  $R_{it}$  is  $\frac{g_{A_i}}{g_{R_i}} = r_i$ .

Prior literature, mostly outside of economics, has often referred to this term as the “returns to research”. We avoid this terminology for two reasons. First, there are several distinct notions of returns:  $\lambda_i$  measures the (static) returns to research effort in producing the flow of new technology;  $1 - \beta_i$  measures the (dynamic) returns to the level of technology in producing the flow of new technology; and  $r_i$  measures the returns (along the BGP) in the level of technology to the level of research effort. Second,  $r_i$  only measures direct, not total, returns: with spillovers,  $\frac{d \log A_i}{d \log R_i} \neq r_i$ .

ably, the same feedback matrix also determines whether technological feedback instead becomes sufficiently strong to generate hyperbolic growth, as the next proposition shows.

**General case.** From here we state the central mathematical result, which we will subsequently apply to a variety of growth environments.

**Proposition 1** (Technological Feedback Loops). *For the innovation network (20), the spectral radius  $\rho(\mathbf{F}^A)$  of the technological feedback matrix (23) partitions the dynamics into three cases.*

1. *If the spectral radius is less than one,  $\rho(\mathbf{F}^A) < 1$ , the system is non-explosive and there exists a balanced growth path where*

$$\mathbf{g}_A^{\text{BGP}} = (\mathbf{I} - \mathbf{F}^A)^{-1} \mathbf{r} \mathbf{g}_R \quad (24)$$

2. *If the spectral radius equals one,  $\rho(\mathbf{F}^A) = 1$ , the system is non-explosive, but no BGP exists; if the network is additionally irreducible, then the system is bounded by double-exponential growth.*
3. *If the spectral radius is greater than one,  $\rho(\mathbf{F}^A) > 1$ , the system exhibits hyperbolic growth, with at least one technology going to infinity in finite time.*

*In particular, there is hyperbolic growth if and only if  $\rho(\mathbf{F}^A) > 1$ .*

*Proof.* See Appendix A. □

Proposition 1 reduces the long-run behavior of the innovation network (20) to a single scalar, the spectral radius  $\rho(\mathbf{F}^A)$ , which intuitively summarizes the strength of technological feedback loops through the network. There is hyperbolic growth if and only if it is greater than one, generalizing the single-sector explosion condition heuristically introduced in the introduction that, for some  $y \in \mathbb{R}_{++}$ ,  $\frac{dy}{dt} = y^{1-\beta}$  explodes if and only if the exponent  $(1 - \beta)$  is greater than 1.<sup>11</sup> Stronger diminishing returns  $\beta_i$  weakly decrease  $\rho(\mathbf{F}^A)$ , pushing the system away from explosive growth, thus generalizing lesson one of section 2.1; stronger spillovers  $\phi_{i,j}$  weakly increase  $\rho(\mathbf{F}^A)$ , pushing the system toward explosive growth, generalizing lesson two of section 2.2.<sup>12</sup>

<sup>11</sup>In a related innovation-network setting, Liu and Ma (2024, Appendix B.9) also obtain a condition on the spectral radius of the innovation network for explosive growth. In another related contribution, Growiec (2007) shows that balanced growth requires the Jacobian of the system to be singular.

<sup>12</sup>The proposition states that “at least one” variable goes to infinity at some finite time  $T$ . In fact, all variables in the same strongly connected component of the network as any exploding technology “want to explode”, but some may not go to infinity only because the system is ill-defined after time  $T$ . Under natural continuations of the system, all such variables do in fact go to infinity: e.g. replace each spillover input  $A_j$  by  $A_j^* \equiv \min\{A_j, \overline{M}\}$ , for constant  $\overline{M}$ , and take  $\overline{M} \rightarrow \infty$ . We thank Philip Trammell for highlighting this point.

When  $\rho(\mathbf{F}^A) < 1$ , the technological Leontief inverse is defined as  $\Psi_A \equiv (\mathbf{I} - \mathbf{F}^A)^{-1}$ . The balanced growth path in (24) can then be written as  $\mathbf{g}_A^{\text{BGP}} = \Psi_A \mathbf{arg}_R$ . The Leontief inverse captures the amplification of technological progress through repeated rounds of technological feedback, as can be seen by expanding it as  $\Psi_A = (\mathbf{I} - \mathbf{F}^A)^{-1} = \mathbf{I} + \mathbf{F}^A + (\mathbf{F}^A)^2 + (\mathbf{F}^A)^3 + \dots$ . Successive terms capture increasingly indirect feedback effects:  $\mathbf{F}^A$  captures the first round of spillovers through the network,  $(\mathbf{F}^A)^2$  captures spillovers operating through one intermediate sector, and so on. Recalling that  $F_{i,j}^A = \frac{\phi_{i,j}}{\beta_i}$  for off-diagonal elements, we can see that entries of the Leontief inverse  $\Psi_A$  are weakly increasing with spillover terms  $\phi_{i,j}$  and decreasing with the strength of diminishing returns to research  $\beta_i$ . In the case of hyperbolic growth, the series above diverges; stronger spillovers or weaker diminishing returns make this more likely.

The technological feedback matrix  $\mathbf{F}^A$  therefore plays a dual role. Its spectral radius determines whether technological feedback is strong enough to generate hyperbolic growth, and in the case  $\rho(\mathbf{F}^A) < 1$ , its Leontief inverse determines the BGP growth rate.

The BGP formula (24) also nests the canonical semi-endogenous balanced growth path presented in Jones (1995). Turn off spillovers,  $\phi_{i,j} = 0$  for  $i \neq j$  (so that the Leontief inverse is the identity matrix,  $\Psi_A = \mathbf{I}$ ), and suppose research inputs are proportional to population, with population growth vectorized as  $\mathbf{n} \equiv n\mathbf{1}$ . Then the vector of balanced growth rates across technological sectors, assuming it exists, is  $\mathbf{g}_A^{\text{BGP}} = \mathbf{r}\mathbf{n}$ .

### 3.2 Economic and technological feedback loops

Now we embed the innovation network of (20) into a broader economic environment. Specifically, instead of fixing the growth rate of research inputs  $g_{R_i}$  as above, we endogenize this growth rate through a lab equipment model. Technological progress increases final output, which allows for additional investment, feeding back into technological progress.

Specifically, consider the following system:

$$\dot{A}_i = v_i [K_i^{\gamma_i} L_i^{1-\gamma_i}]^{\lambda_i} A_i^{1-\beta_i} \prod_{j \neq i} A_j^{\phi_{i,j}} \quad (25)$$

$$\dot{K} = aY - \delta K \quad (26)$$

$$Y = \bar{A} K_Y^\alpha L_Y^{1-\alpha} \quad (27)$$

$$\bar{A} = \prod_i A_i^{\tau_i} \quad (28)$$

$$K = K_Y + \sum_i K_i \quad (29)$$

$$L = L_Y + \sum_i L_i \quad (30)$$

Here,  $a > 0$  is a constant savings rate; the shares of capital and labor allocated to each technology and to output remain constant; and sectoral labor grows exogenously at a common constant rate  $n > 0$ , so  $\mathbf{n} = n\mathbf{1}$ .

Compared to the system studied above, the ideas production function here (25) explicitly specifies that capital and labor are inputs, a la Cobb-Douglas, into innovation. It also introduces a new law of motion: for capital, (26), featuring investment financed from output  $Y$ . Different technologies are combined into final-goods TFP  $\bar{A}$  (with constant returns to scale,  $\sum_i \tau_i = 1$ ) as specified in (28). Equations (29) and (30) specify resource constraints for capital and labor.

We will make use of an  $N \times N$  economic feedback matrix,  $\mathbf{F}^Y$ :

$$F_{i,j}^Y = \tau_j \cdot \frac{1}{1-\alpha} \cdot \gamma_i \cdot r_i \quad (31)$$

Entries in this matrix describe how technological progress in sector  $j$  feeds back into technological progress in sector  $i$  through an economic feedback loop. Higher technology in sector  $j$ ,  $A_j$ , raises output  $Y$  (through  $\tau_j$ ); the economic feedback loop between output and capital  $K$  amplifies this (through  $\frac{1}{1-\alpha}$ ); the resulting higher capital is used in part as a research input in sector  $i$  (through  $\gamma_i$ ); and greater research inputs raise technology in sector  $i$ ,  $A_i$  (through  $r_i$ ).

The economic feedback matrix is the economic analog of the technological feedback matrix,  $\mathbf{F}^A$ . In matrix form, it is written  $\mathbf{F}^Y \equiv \frac{1}{1-\alpha}[\mathbf{r}\boldsymbol{\gamma}]\boldsymbol{\tau}'$ , where  $\boldsymbol{\gamma}$  is the vector of  $\gamma_i$  exponents,  $\boldsymbol{\tau}$  is the vector of  $\tau_i$  exponents, and  $\mathbf{r} = \text{diag}(r_1, \dots, r_N)$ .

The next proposition describes growth dynamics in the above system, where there are both economic and technological feedback loops. It is analogous to Proposition 1, which studied technological feedback loops alone, but replaces the technological feedback matrix  $\mathbf{F}^A$  with the sum of  $\mathbf{F}^A$  and the economic feedback matrix  $\mathbf{F}^Y$ .

**Proposition 2** (Economic and Technological Feedback Loops). *Consider the economy described in (25)-(30). The spectral radius of  $\mathbf{F}^A + \mathbf{F}^Y$  partitions the growth system into three cases:*

1. *If the spectral radius is less than one,  $\rho(\mathbf{F}^A + \mathbf{F}^Y) < 1$ , then the system is non-explosive and there exists a balanced growth path where:*

$$\mathbf{g}_A^{\text{BGP}} = [\mathbf{I} - (\mathbf{F}^A + \mathbf{F}^Y)]^{-1} \mathbf{r}\mathbf{n} \quad (32)$$

*where  $\mathbf{F}^A$  and  $\mathbf{F}^Y$  are the technological and economic feedback matrices defined in (23) and (31), while  $\mathbf{r}$  is defined in (22).*

2. *If the spectral radius equals one,  $\rho(\mathbf{F}^A + \mathbf{F}^Y) = 1$ , the system is non-explosive, but no*

*BGP exists; if the network is additionally irreducible, then the system is bounded by double-exponential growth.*

3. *If the spectral radius is greater than one,  $\rho(\mathbf{F}^A + \mathbf{F}^Y) > 1$ , the system exhibits hyperbolic growth, with at least one variable going to infinity in finite time.*

*In particular, there is hyperbolic growth if and only if  $\rho(\mathbf{F}^A + \mathbf{F}^Y) > 1$ .*

*Proof.* See Appendix B.1. □

This proposition says that there is hyperbolic growth if the spectral radius of the sum of the technological and economic feedback loops is greater than one: that is, if the sum of the two kinds of feedback loops is strong enough. Earlier, we saw how strengthening technological feedback loops (larger spillovers  $\phi_{i,j}$  or weaker diminishing returns  $\beta_i$ ) made explosive growth more likely, through  $\mathbf{F}^A$ . Now, we see that stronger economic feedback loops make explosive growth more likely as well, through  $\mathbf{F}^Y$ : higher capital shares in production  $\alpha$  or in innovation  $\gamma_i$  both make explosive growth more likely. This generalizes lesson three of section 2.3. The formal proof of the proposition in Appendix B.1 is structurally similar to the proof of Proposition 1 in Appendix A.

In the case of balanced growth, the Leontief inverse  $\Psi_{A,Y} \equiv [\mathbf{I} - (\mathbf{F}^A + \mathbf{F}^Y)]^{-1}$  captures the amplification of technological progress through repeated rounds of technological and economic feedback. It can be written as  $\Psi_{A,Y} = \mathbf{I} + (\mathbf{F}^A + \mathbf{F}^Y) + (\mathbf{F}^A + \mathbf{F}^Y)^2 + \dots$ . We can observe, via the higher-order terms, that there is complementarity between economic and technological feedback.

In this model we assume that savings rates are constant (but may take arbitrary positive values). Of course, optimal savings paths will respond endogenously to changes in the growth rate. If the saving rate varies but stays boundedly away from zero, the explosion condition is unchanged. The substantive question is therefore whether savings could, endogenously, become sufficiently small to avert an explosion. Trammell and Korinek (2026) consider a related setting and show that, under standard preferences, the Euler equation prevents declining saving from eliminating superexponential growth. Their argument takes the growth of capital productivity as given, whereas technological progress in our model depends on capital accumulation. We leave a formal treatment of optimal saving to future work.

The law of motion for capital underlying Proposition 2, equation (26), is qualitatively different from the law of motion for technology studied in Proposition 1, because of the linear drag imposed by depreciation  $\delta$ . The proof of the proposition demonstrates that the depreciation term does not impact whether the system exhibits explosive growth. Jones (2026) demonstrates the same in a simpler economic environment.

## 4 AI and automation

In this section, we incorporate AI-driven automation and show that such automation triggers new feedback loops or strengthens existing ones.

### 4.1 Automation

Here we adapt the automation framework of Zeira (1998) and Acemoglu and Restrepo (2018), where *effective labor* in each sector,  $\hat{L}_i$ , is a Cobb-Douglas aggregation of outputs from  $T$  different tasks. In particular, we assume that *effective labor* working in a sector  $i$  is given by  $\hat{L}_i = p_i T \prod_q X_{i,q}^{1/T}$ , where  $p_i$  is a sector-specific productivity constant. If task  $q$  is not automated, it is produced with an amount of labor,  $X_{i,q} = L_{i,q}$ . Otherwise, it is produced by AI labor,  $X_{i,q} = L_{AI,i,q}$ , which is described momentarily. Let  $f_i$  denote the fraction of tasks in sector  $i$  which are automatable by AI; assuming that compute capital is sufficiently abundant, all tasks which are automatable are automated. Then, it is optimal to spread AI labor equally across automated tasks and to spread human labor equally across non-automated tasks. Effective labor is thus given by:

$$\hat{L}_i(L_{AI,i}, L_i) = L_{AI,i}^{f_i} L_i^{1-f_i} \tilde{p}_i(f_i) \quad (33)$$

where  $\tilde{p}_i(f_i) \equiv \frac{p_i}{f_i^{f_i} (1-f_i)^{1-f_i}}$ , which is an unimportant constant for a given level of automation and does not affect explosive dynamics. Importantly, we here treat the level of automation  $f_i$  as exogenously given, and we evaluate the growth implications for a given level of task automation.

### 4.2 AI-driven automation

Sector- $i$  AI labor,  $L_{AI,i}$ , is determined both by compute used in that sector and by the level of software capabilities in the economy. We present the notation and then explain.

AI labor in sector  $i$  is:

$$L_{AI,i} = \underbrace{C_i}_{\text{rival}} \times \underbrace{\prod_{j \in I} A_j^{\sigma_j}}_{\text{nonrival}} \quad (34)$$

Compute (or “hardware”) is a particular kind of capital required to perform AI labor. In particular, it reflects compute used for AI inference. We denote the quantity used in sector  $i$  as  $C_i$ . Note this is a rivalrous good: use in one sector precludes use in another

sector. Total inference compute across the economy is denoted  $C$ .

The level of software capabilities in the economy – also frequently termed “algorithmic progress” – is nonrivalrous and determined by a composite of ideas from different research sectors,  $\prod_{i \in I} A_i^{\sigma_i}$ . We assume  $\sigma_i \geq 0$ , allowing  $\sigma_i = 0$  if a sector does not contribute to software progress, and we do not require  $\sum_{i \in I} \sigma_i = 1$  since standard replication arguments for constant returns to scale do not apply to nonrivalrous technologies. Examples of algorithmic progress include the original invention of the neural network; the modern deep learning paradigm; and the transformer.

Thus, (34) takes software capabilities to raise the productivity of a given quantity of compute. This choice to model AI progress as arising from both inference compute scaling ( $C$ ) and software progress ( $\prod_j A_j^{\sigma_j}$ ) is inspired by the crucial stylized fact that AI progress has been driven both by increasing compute inputs and also by algorithmic advances (Huet al., 2024).

### 4.3 Hyperbolic growth under automation of research

In this subsection, we allow for AI-driven automation of the innovation network. We set aside economic feedback loops.

Assume that the only rival input to research is effective labor, combining both human labor and AI effective labor according to equation (33).<sup>13</sup> Therefore the technology laws of motion become

$$\begin{aligned}
\dot{A}_i &= v_i \hat{L}_i^{\lambda_i} A_i^{1-\beta_i} \prod_{j \neq i} A_j^{\phi_{i,j}} \\
&= v_i \left[ \tilde{p}_i(f_i) L_{AI,i}^{f_i} L_i^{1-f_i} \right]^{\lambda_i} A_i^{1-\beta_i} \prod_{j \neq i} A_j^{\phi_{i,j}} \\
&\propto \underbrace{A_i^{1-\beta_i}}_{\text{diminishing returns}} \times \underbrace{[C_i^{f_i} L_i^{1-f_i}]^{\lambda_i}}_{\text{rival inputs}} \times \underbrace{\prod_{j \in I} A_j^{f_i \lambda_i \sigma_j}}_{\text{AI feedback}} \times \underbrace{\prod_{j \neq i} A_j^{\phi_{i,j}}}_{\text{direct spillovers}} \quad (35)
\end{aligned}$$

The innovation network with AI-driven automation features two kinds of technological spillovers. First, “direct” spillovers as before, when  $\phi_{i,j} > 0$ . Second, there are “indirect” spillovers when a sector is at least partially automated by AI, through the technologies that feed into software,  $\sigma_j > 0$ .

As a baseline, we assume that direct technological spillovers are non-existent ( $\phi_{i,j} = 0$ ). Thus sectoral spillovers occur exclusively indirectly, through research automation: a

<sup>13</sup>Allowing for non-AI capital  $K_i$  would introduce further economic feedback loops.

positive automation shock to a sector that feeds into AI progress accelerates research in all other sectors that have any automation.

We can map the innovation network without automation, (25), into the innovation network here, (35), by matching terms.<sup>14</sup> This mapping is summarized in table 1.

Table 1: Mapping between innovation network models

| Without automation (25) | With automation (35)                                                   | Description                       |
|-------------------------|------------------------------------------------------------------------|-----------------------------------|
| $\beta_i$               | $\beta_i - f_i \lambda_i \sigma_i$                                     | Diminishing returns               |
| $\phi_{i,j}$            | $\underbrace{\phi_{i,j}}_{\text{assume } =0} + f_i \lambda_i \sigma_j$ | Automation spillovers             |
| $1 - \gamma_i$          | $1 - f_i$                                                              | Non-automated research task share |
| $\lambda_i$             | $\lambda_i$                                                            | Parallelization penalty           |

From this mapping, we make several observations:

- Diminishing returns to research within a sector,  $\beta_i$ , are directly offset by the ability for AI to contribute to research in that sector. Introducing automated research is isomorphic to offsetting diminishing returns.
- Automation increases technological spillovers. Note that for progress in  $j$  to spill over into progress in  $i$ , it is required that *both* sector  $i$  is partially automated ( $f_i > 0$ ) and that sector  $j$  is relevant to AI progress ( $\sigma_j > 0$ ). Therefore, a ‘bilateral’ feedback loop exists between  $i$  and  $j$  if both  $f$  and  $\sigma$  are greater than zero for both  $i$  and  $j$ . An ‘indirect’ feedback loop exists if there is a chain of technologies with positive  $f$  and  $\sigma$  such that progress in one sector eventually reinforces itself such that progress in one sector accelerates progress in a second, which accelerates a third, and so on until the original sector is itself accelerated.

The isomorphism between the two models means that Proposition 1, which was derived without automation, can be applied directly to this system.

The argument used to prove Proposition 1 also applies to this system, with minimal modification, as shown in Appendix B.2. This yields the following explosion condition:

<sup>14</sup>In this mapping with (25), compute  $C_i$  is mapped to capital  $K_i$ .

**Corollary 1.** *The automation-augmented technology system (equation (35)) explodes in finite time iff*

$$\sum_{i \in I} r_i f_i \sigma_i > 1$$

*Proof.* See Appendix B.2. □

In the case of a balanced growth path, meanwhile, technology growth is  $\mathbf{g}_A^{\text{BGP}} = \hat{\Psi}_A \hat{\mathbf{r}} \hat{\mathbf{g}}_R$ , where  $\hat{g}_{R_i} = (1 - f_i) \times n + f_i \times g_{C_i}$ ,  $\hat{r}_i \equiv \frac{\lambda_i}{\beta_i - f_i \lambda_i \sigma_i}$ ,  $\hat{F}_{i,j}^A \equiv \frac{f_i \lambda_i \sigma_j}{\beta_i - f_i \lambda_i \sigma_i}$  for  $i \neq j$ ,  $\hat{F}_{i,i}^A = 0$ ,  $\hat{\Psi}_A \equiv (\mathbf{I} - \hat{\mathbf{F}}^A)^{-1}$ , and  $\hat{\mathbf{r}} \equiv \text{diag}(\hat{r}_1, \dots, \hat{r}_N)$ .<sup>15</sup>

## 4.4 Hyperbolic growth: Automated production and research

Now we embed the innovation network with AI-driven automation of the last subsection into an economy with economic feedback loops, and we allow AI to additionally automate production. Consider the model:

$$\dot{A}_i = v_i (\hat{L}_i)^{\lambda_i} A_i^{1-\beta_i} \prod_{j \neq i} A_j^{\phi_{i,j}} \quad (36)$$

$$\bar{A} = \prod_i A_i^{\tau_i} \quad (37)$$

$$Y = \bar{A} K^\alpha (\hat{L}_Y)^{1-\alpha} \quad (38)$$

$$\hat{L}_i = L_i^{1-f_i} L_{\text{AI},i}^{f_i} \tilde{p}_i \quad (39)$$

$$L_{\text{AI},i} = C_i \prod_j A_j^{\sigma_j} \quad (40)$$

$$\dot{K} = a_K Y - \delta_K K \quad (41)$$

$$\dot{C} = a_C Y - \delta_C C \quad (42)$$

$$C = C_Y + \sum_i C_i \quad (43)$$

$$L = L_Y + \sum_i L_i \quad (44)$$

where investment rates  $0 < a_K, a_C < 1$  sum to less than one and allocations of compute and labor across research sectors are constant over time (i.e.,  $C_i/C$  and  $L_i/L$  are constant). Note that effective labor now automates goods production (38), with  $f_Y \in [0, 1)$  now reflecting the share of tasks that are automated in goods production.<sup>16</sup>

<sup>15</sup>Note that the explosion condition is written in terms of the unadjusted  $r_i = \frac{\lambda_i}{\beta_i}$ , not in terms of  $\hat{r}_i = \frac{\lambda_i}{\beta_i - f_i \lambda_i \sigma_i}$ .

<sup>16</sup>We assume  $f_Y < 1$  throughout, while research sectors may be fully automated. At  $f_Y = 1$ , whether growth explodes can additionally depend on initial conditions ( $K_0$ ) and depreciation.

Once again, this model can be mapped to an earlier model without automation, so that we can reuse an earlier proposition. Table 2 presents how the model with economic and technological feedback loops but without automation, (25)-(30), maps to the model here with automation. From this, we can see that introducing automation into the production side of the model (in addition to the technological side) is equivalent to: increasing the capital share of production by  $f_Y(1 - \alpha)$ ; and increasing the contributions of each technology to the production of final goods by  $(1 - \alpha)\sigma_i f_Y$ .

**Table 2:** Mapping between innovation network models with economic feedback loops: without and with automation

| Without automation<br>(25)-(30) | With automation<br>(36)-(44)                                         | Description                           |
|---------------------------------|----------------------------------------------------------------------|---------------------------------------|
| $\alpha$                        | $\alpha + f_Y(1 - \alpha)$                                           | Capital share of output               |
| $\tau_i$                        | $\tau_i + (1 - \alpha)\sigma_i f_Y$                                  | Technological contributions to output |
| $\beta_i$                       | $\beta_i - f_i \lambda_i \sigma_i$                                   | Diminishing returns                   |
| $\phi_{i,j}$                    | $\underbrace{\phi_{i,j}}_{\text{assume}=0} + f_i \lambda_i \sigma_j$ | Automation spillovers                 |
| $1 - \gamma_i$                  | $1 - f_i$                                                            | Non-automated research task share     |
| $\lambda_i$                     | $\lambda_i$                                                          | Parallelization penalty               |

Applying Proposition 2, we have the following result:

**Corollary 2.** Consider the growth model with economic feedback loops, (36)-(44), and assume no direct spillovers,  $\phi_{i,j} = 0$  for all  $i \neq j$ . The system explodes in finite time iff:

$$f_Y + \sum_i r_i f_i \left( \frac{\tau_i}{1 - \alpha} + \sigma_i \right) > 1 \quad (45)$$

*Proof.* See Appendix B.2. □

In the case of a balanced growth path, meanwhile, technology growth is

$$\mathbf{g}_A^{\text{BGP}} = \hat{\Psi}_{A,Y} \hat{\mathbf{r}} \mathbf{n} \quad (46)$$

where

$$\hat{\mathbf{F}}^Y \equiv [\hat{\mathbf{r}}\mathbf{f}] \left[ \frac{1}{1-f_Y} \left( \frac{1}{1-\alpha} \tau + f_Y \sigma \right) \right]' \quad (47)$$

$$\hat{\Psi}_{A,Y} \equiv [\mathbf{I} - (\hat{\mathbf{F}}^A + \hat{\mathbf{F}}^Y)]^{-1} \quad (48)$$

Here,  $\hat{\mathbf{r}}$  and  $\hat{\mathbf{F}}^A$  are defined as above,  $\mathbf{f}$  is the vector of automation shares in each technology sector, and  $\sigma \equiv (\sigma_1, \dots, \sigma_N)'$ .

## 5 Application: A simple integrated AI-economy model

### 5.1 Setup and explosion condition

Here we integrate a model of AI progress into an economic environment, applying the general model of section 4, which is simple enough to be calibrated to data. As in previous sections, we assume that AI contributes to effective labor, replacing human labor in some fraction of tasks in each sector. The central force continues to be that AI progress stems from both better algorithms and better computing hardware. Improved computing hardware allows us to run more computations for the same amount of capital investment, while improved algorithms make AI more capable of completing relevant tasks.

**Setup.** We present the equations of the model before describing them in words:

$$Y = A \hat{L}_Y^{1-\alpha} K^\alpha \quad (49)$$

$$\hat{L}_i = L_i^{1-f_i} \cdot (L_{AI,i})^{f_i} \cdot \tilde{p}_i \quad (50)$$

$$L_{AI,i} = C_i S \quad (51)$$

$$\dot{A} = v_A \hat{L}_A^{\lambda_A} A^{1-\beta_A} \quad (52)$$

$$\dot{S} = v_S \hat{L}_S^{\lambda_S} S^{1-\beta_S} \quad (53)$$

$$\dot{H} = v_H \hat{L}_H^{\lambda_H} H^{1-\beta_H} \quad (54)$$

$$\dot{C} = a_C H Y - \delta_C C \quad (55)$$

$$\dot{K} = a_K Y - \delta_K K \quad (56)$$

$$C = C_Y + C_A + C_S + C_H \quad (57)$$

$$L = L_Y + L_A + L_S + L_H \quad (58)$$

Final goods  $Y$  are produced with effective labor  $\hat{L}_Y$  and traditional capital  $K$ . Effective labor is a composite of a bundle of tasks, aggregated with an elasticity of substitution equal to one, where a fraction  $f$  of tasks are performed by AI labor  $L_{AI}$  and the remainder by human labor  $L$ . AI labor itself is hardware capital  $C$  (“compute”) scaled by software efficiency  $S$ .

There are three independent technological processes: ideas used for final production  $A$  (abusing terms, “general TFP”); software  $S$ ; and hardware quality  $H$ . Traditional capital  $K$  and compute  $C$  accumulate a la Solow with investment rates  $a_K$  and  $a_C$ ; there is investment-specific technical change for compute investment in the form of hardware quality improvements  $H$ . Human labor  $L$  grows exponentially at rate  $n > 0$ .

We deliberately omit a number of additional components of the training process here to emphasize key feedback loops. For example, our model exclusively focuses on inference without modeling the relationship between pretraining compute and AI capabilities. Erdil et al. (2025) consider a much richer AI-economic model, though such a framework makes it impossible to cleanly isolate the key feedback loops of interest.

**Explosive growth condition.** Just as in section 4.4, we can find the conditions for balanced and explosive growth by recognizing from equations (55) and (56) that a balanced growth path requires  $g_K = g_C - g_H = g_Y$ . Then we can use the equations for output and effective labor to solve for the balanced growth path of technologies as a function of fundamentals of the model. This is the same balanced growth condition for AI capabilities as in the general model from section 4.4. Further, note that we only have one technology – TFP – that feeds directly into increasing output productivity.

Therefore, we can see that the system described in equations (49)-(58) is a three technology version of the general model from equations (36)-(44), where: technological contributions to output are dictated by  $\tau_S = \tau_H = 0$  and  $\tau_A = 1$ ; and technological contributions to AI progress are dictated by  $\sigma_S = \sigma_H = 1$  and  $\sigma_A = 0$ .<sup>17</sup>

Given that the AI-economic model developed here is a specific case of the general model, we can simply apply Corollary 2 to derive the hyperbolic growth condition that was first presented in equation (1) in the introduction:

---

<sup>17</sup>The only difference between this applied model and the general model from section 4.4 is that one of the technologies – hardware quality – scales compute accumulation rather than AI capabilities directly. That is, it is a form of investment-specific technical change. Ultimately this does not affect the hyperbolic growth condition. To see this, one can show that the balanced growth of  $L_{AI}$  under either (42) or (55) is  $g_{L_{AI}} = g_Y + g_H + g_S$ . Thus, the equations determining balanced growth are unchanged, and hence so is the threshold for hyperbolic growth.

**Corollary 3.** *The AI-economic model (described in equations (49)-(58)) explodes in finite time iff*

$$f_Y + f_S r_S + f_H r_H + \frac{1}{1 - \alpha} f_A r_A > 1 \quad (59)$$

## 5.2 Calibration: Explosive vs. balanced growth

We now turn to calibrating the relevant terms from equation (59). In table 3 we report estimates of diminishing returns in software, hardware, and aggregate TFP. Both software and (in particular) hardware research face significantly less diminishing returns than aggregate TFP. That is, ideas are getting harder to find in all sectors, but less so in software and particularly hardware. We discuss limitations of this calibration in section 5.4.

**Table 3:** Calibration of technological parameters

| Term                                                   | Parameter                              | Estimate | Source                                          |
|--------------------------------------------------------|----------------------------------------|----------|-------------------------------------------------|
| Labor share                                            | $1 - \alpha$                           | 0.6      | Pre-AI ( $f_Y = 0$ ) labor share                |
| Inverse of diminishing returns to research in TFP      | $r_A \equiv \frac{\lambda_A}{\beta_A}$ | 0.32     | <a href="#">Bloom et al. (2020)</a>             |
| Inverse of diminishing returns to research in software | $r_S \equiv \frac{\lambda_S}{\beta_S}$ | 1        | <a href="#">Erdil, Besiroglu, and Ho (2024)</a> |
| Inverse of diminishing returns to research in hardware | $r_H \equiv \frac{\lambda_H}{\beta_H}$ | 5        | <a href="#">Bloom et al. (2020)</a>             |

Calibrating (59) using table 3, the condition becomes:

$$f_Y + 1 \cdot f_S + 5 \cdot f_H + 0.53 \cdot f_A > 1 \quad (60)$$

A key takeaway of the exercise is that automating hardware research – increasing  $f_H$  – has the highest impact of automation across any sector, since diminishing returns to research are so low in that sector, or conversely that  $r_H = 5$  is so high. Automating one hardware research task offers about the same increase in the left-hand side of (60) as five tasks in software or final goods production and the same as ten tasks in the general TFP sector.

Condition (60) allows us to calculate what fraction of each sector needs to be automated to generate explosive growth. Table 4 performs this exercise, where each row considers automating a different set of sectors an equal amount with zero automation in other sectors. Column two solves for how much automation is required to achieve a doubling of output growth on the balanced growth path; here we assume labor growth is uniform across sectors. Column three, meanwhile, solves for the level of automation required to achieve the hyperbolic growth threshold, under the same assumption of equal automation across sectors.

Table 4: Automation required for hyperbolic growth or doubling growth

| Automated sectors | Automation threshold        |              |
|-------------------|-----------------------------|--------------|
|                   | $2 \times g_Y^{\text{BGP}}$ | Hyperbolic   |
| $S$               | –                           | $\sim 100\%$ |
| $H$               | –                           | 20%          |
| $S, H$            | –                           | 17%          |
| $H, Y$            | 10%                         | 17%          |
| $S, Y$            | 16%                         | 50%          |
| $S, H, Y$         | 9%                          | 14%          |
| $S, H, A, Y$      | 7%                          | <b>13%</b>   |

*Note:* Here we assume equal automation across automated sectors. The middle column displays the level of automation required such that output growth on the balanced growth path doubles relative to zero automation. The right column displays the level of automation such that there is hyperbolic growth. Doubling output growth is not possible in the case of the first three rows.

The table reports, in particular, an economy-wide automation threshold of 13%: if all sectors across software, hardware research, general research, and goods production are automated 13%, then the economy is on a hyperbolic path. We can also see that, in that case, the automation required to double the balanced growth path rate of growth (7%) is roughly half the automation necessary to achieve fully explosive growth (13%). That is, the balanced growth path is initially very slow to respond to changes in automation, and then changes quickly. This is the result of the multiplication of spillovers through the Leontief inverse.

These quantitative results underscore two points. First is the importance of amplifi-

cation of feedback loops. Under the simple model developed above, higher levels of automation – dialing up feedback loops – quickly translate into rapid changes in the growth path. Second is the emphasis that these are changes in the asymptotic balanced growth path. In reality it may take some time to converge to such a path and further, even under a hyperbolic growth path it may still take significant time to actually see radical changes in growth rates; hyperbolic growth does not necessarily imply transformative growth in the short run. One can see this intuitively by recognizing that, from the viewpoint of the long sweep of human history, the current 2% annual growth is part of a long-run super-exponential trend (Hanson, 2000; Roodman, 2020).

That said, empirically the recent growth rates of productivity in software and hardware have been extraordinarily fast, and so it is also plausible that the transition to a new balanced growth path or hyperbolic acceleration happens extremely quickly. We turn to quantifying these considerations in the next subsection.

### 5.3 Simulation: The effect of an automation shock over time

We now simulate an entire time path for technology and output following a one-time increase in the share of automated tasks, and (when the explosion condition is satisfied) calculate the singularity date implied by the model. Specifically, starting from zero automation, we simulate the model after a jump in sectoral automation rates: for  $i \in \{S, H, A, Y\}$ , we assume  $f_{i,t} = 0$  for  $t < 0$ , and  $f_{i,t} = f_i$  for  $t \geq 0$ .

$$f_{i,t} = \begin{cases} 0 & \text{if } t < 0 \\ f_i & \text{if } t \geq 0 \end{cases} \quad (61)$$

**Calibration.** The simulation requires introducing additional quantitative assumptions beyond those in the previous subsections.

We assume the economy starts on a balanced growth path with zero automation, for which  $g_{A_i}^{\text{BGP}} = r_i n_i$ . Unlike in the previous subsection, we now assume growth in labor inputs can vary across sectors. With the BGP relation, we can back out researcher growth in each sector ( $n_H$  and  $n_S$ ) by combining the ideas-getting-harder-to-find parameters in table 3 together with measured, current technological growth rates in software and hardware reported by Epoch AI (2026):  $g_{S,2026} = 110\%$  and  $g_{H,2026} = 37\%$ . Together these imply that the number of researchers in the hardware sector has been growing at  $n_H = 7.4\%$  and in the software sector at  $n_S = 110\%$  per year. Reassuringly, these numbers are close to direct

measurements of these parameters.<sup>18,19</sup> Additionally, we take from Bloom et al. (2020) that aggregate effective research effort has been growing at a rate of  $n_A = 4\%$  annually. In the output sector, we assume that the growth rate of workers is  $n_Y = 1\%$ .

In the task model, an automation shock produces two growth effects: reallocating human workers into a narrower range of tasks, and increasing the elasticity of effective labor toward an endogenously accumulating factor. The first causes an immediate jump in growth rates; the second shapes the long-run path of variables. We focus on the second and therefore assume no labor reallocation after the shock. Mathematically, this amounts to setting  $\tilde{p}_i(f_i) = 1$  for all  $i$  in the effective labor function  $\hat{L}_i = \tilde{p}_i(f_i) L_i^{1-f_i} L_{AI,i}^{f_i}$ .

In addition to these key calibrations, we assume:

- Initial sectoral compute allocations are chosen so that  $L_{AI,i,0} = L_{i,0}$ . Therefore,  $L_{i,0}^{1-f_i} L_{AI,i,0}^{f_i} = L_{i,0}$  for any  $f_i$ , and effective labor is unchanged by the shock. This restriction implies technology growth rates do not jump at  $t = 0$ , so that all subsequent acceleration comes from endogenous accumulation and technological progress.
- Depreciation is 7% for conventional capital. Along the no-automation balanced growth path, we assume a standard capital-to-output ratio  $K/Y = 4$ . Since  $g_Y^{\text{BGP}} = g_K^{\text{BGP}}$ , we can arrive at the capital investment rate  $a_K = (g_Y^{\text{BGP}} + \delta_K) \frac{K_0}{Y_0} = 0.41$ . Combined with compute investment, this high savings rate is an implication of calibrating to 2026 data and assuming that the 2026 economy is on a balanced growth path.
- Depreciation is 20% for computing hardware. The initial growth rates of the economy and hardware depreciation pin down the ratio of compute investment to compute stock:  $\frac{a_C H_0 Y_0}{C_0} = g_H^{\text{BGP}} + g_Y^{\text{BGP}} + \delta_C = 0.6$ . Given constant  $a_C$ , this ratio is sufficient to pin down the path of compute accumulation without initial stocks.
- Labor growth in each sector is given by  $L_{i,t} = L_{i,0} e^{n_i t}$ , and  $\frac{C_{i,t}}{C_{i,0}} = \frac{C_t}{C_0}$ . Thus, sectoral compute shares are fixed over time.

Note that, under these assumptions, we do not need to separately calibrate levels or shares; they are pinned down by these assumptions and the observed initial growth rates.<sup>20</sup>

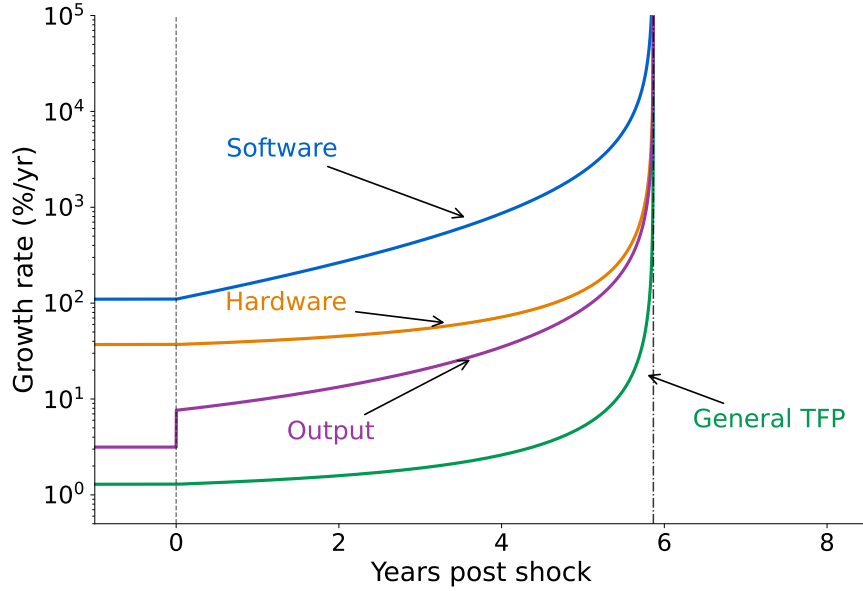
<sup>18</sup>Our calibration of hardware researcher growth,  $n_H = 7.4\%$ , is close to the estimate of Bloom et al. (2020) of 6.8% historical growth in research inputs in the hardware sector. Our calibration of software researcher growth,  $n_S = 110\%$ , is higher than, but of the same order of magnitude as, the estimate by Epoch AI (2026) of growth in OpenAI and Anthropic headcount of roughly  $n_S = 90\%$  per year since GPT-4.

<sup>19</sup>Given assumed – and observed – extremely rapid growth rates of employment in the software sector, sustained growth over many years would, in reality, eventually require unmodeled reallocation of labor out of other sectors into software.

<sup>20</sup>To illustrate, consider software growth pinned down by  $g_{S,t} = v_S \hat{L}_{S,t}^{\lambda_S} S_t^{-\beta_S}$ . Then for  $t < 0$ , assuming growth is balanced we have  $v_S = g_S^{\text{BGP}} L_{S,t}^{-\lambda_S} S_t^{\beta_S}$  where the RHS is constant along the balanced growth path.

**Simulation results.** Figure 6 illustrates the growth rate of our four main variables (software, hardware, general TFP, and output) over time after our main experiment: a jump in automation from  $f = 0\%$  in all sectors to  $f_S = 100\%$  in software and  $f_{-S} = 5\%$  elsewhere. The model predicts a singularity in less than six years.

Figure 6: Automation shock experiment: Full automation of software R&D



*Note:* Growth rates following an increase in automation at  $t = 0$  from zero automation to complete automation of software ( $f_S = 100\%$ ) and minor automation of other sectors ( $f_{-S} = 5\%$ ).

Figure 7 illustrates a different exercise to show how dynamics differ depending on the size of the automation shock. These simulations consider an automation shock that is uniform across sectors ( $f_i = f$  for all  $i$ ). Different lines indicate different degrees of automation,  $f \in \{10\%, 13\%, 15\%, 20\%\}$ . Under our baseline calibration,  $f = 13\%$  is the knife-edge case beyond which there is explosive growth, as previously seen in table 4. With 15% uni-

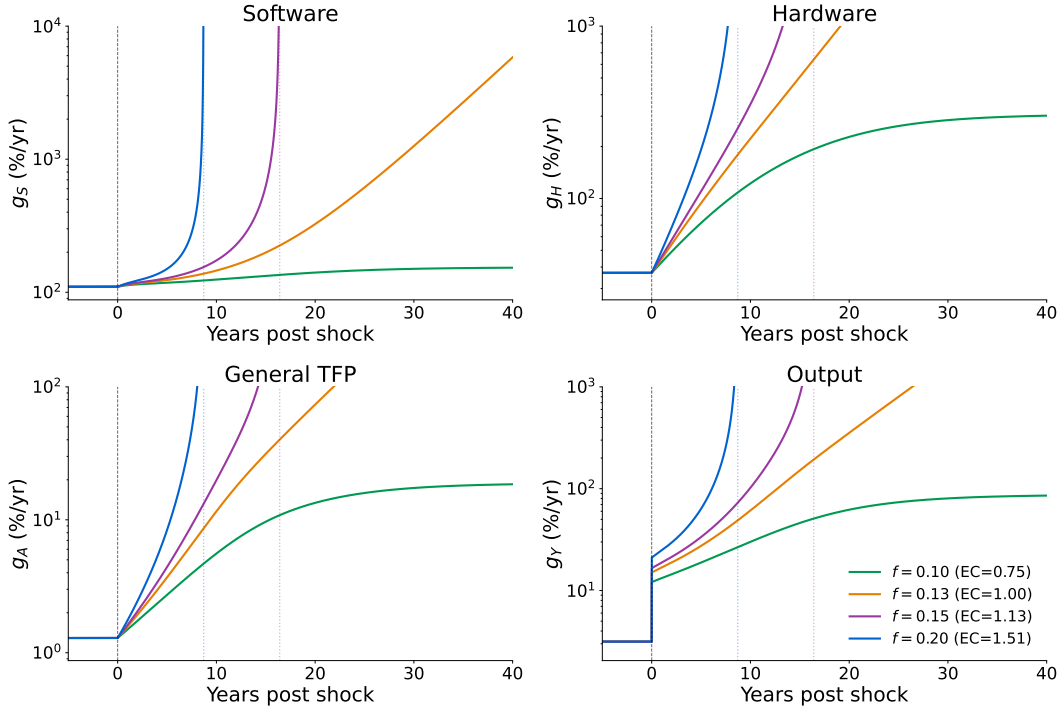
Then, for  $t \geq 0$  we have

$$\begin{aligned} g_{S,t} &= g_{S,0} \left( \frac{L_{S,t}}{L_{S,0}} \right)^{\lambda_S(1-f_S)} \left( \frac{C_{S,t}S_t}{L_{S,0}} \right)^{\lambda_S f_S} \left( \frac{S_t}{S_0} \right)^{-\beta_S} \\ &= g_{S,0} e^{\lambda_S n_S(1-f_S)t} \left( \frac{C_t}{C_0} \frac{S_t}{S_0} \right)^{\lambda_S f_S} \left( \frac{S_t}{S_0} \right)^{-\beta_S} \end{aligned}$$

where the second line follows from the assumptions on sectoral labor growth rates, the effective labor normalization assuring no jump in growth rates, and the stable compute allocations. This can be repeated for all technologies. Note that this means we are also required to calibrate  $\lambda_i$  rather than just  $r_i$ ; to this end we assume no parallelization penalty, so  $\lambda_i = 1$  and  $\beta_i = 1/r_i$  for all  $i$ .

form automation, there is a singularity in 16 years; with 20% uniform automation, there is a singularity within 9 years.

**Figure 7:** Automation shock experiment: Uniform automation across sectors



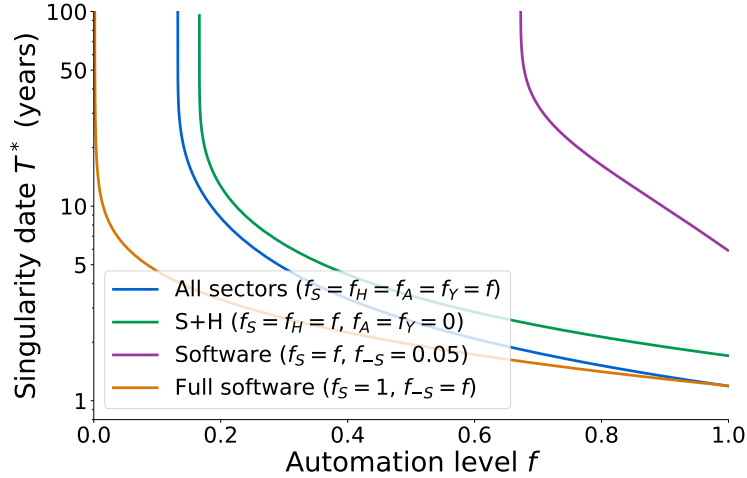
*Note:* Growth rates following an increase in automation at  $t = 0$  from zero automation to a uniform level of automation  $f$  across sectors, starting from a balanced growth path. The legend reports the “explosion condition” value  $EC \equiv f_Y + f_S r_S + f_H r_H + f_A \frac{r_A}{1-\alpha}$ , with  $r$  and  $\alpha$  values from table 3. Per Corollary 3,  $EC > 1$  yields hyperbolic growth. Here, 13% uniform automation is the knife-edge case (orange line); 15% and 20% uniform automation cross into the explosive regime, producing singularities in roughly 16 and 9 years respectively.

Figure 8 plots the singularity date under the above calibration as a function of different automation shocks.<sup>21</sup> The blue line considers uniform automation across sectors, where we see that under full automation ( $f = 1$ ) the singularity arrives just one year after the shock. The green line considers the case where AI only automates the software and hardware industry, where we see that the time to the singularity is longer but not far from the prior case where production ( $Y$ ) and general research ( $A$ ) were also automated. The purple line fixes all non-software sectors at 5% automation, and varies the level of automation in the software sector. The orange line assumes software is fully automated ( $f_S = 1$ ), and

<sup>21</sup>Computationally, we approximate the singularity date by testing when growth in the system exceeds 10,000% per year.

varies uniformly the level of automation in other sectors.

Figure 8: Time to singularity as a function of automation shock



*Note:* Time to singularity (on the y-axis, which is in logs) as a function of the size of the automation shock  $f$  (on the x-axis). Each line considers a different automation shock. “All sectors” increases automation in every sector from 0 to  $f$ . “S+H” increases software and hardware automation from 0 to  $f$ , leaving other sectors unautomated. “Software” increases software automation from 0 to  $f$ , starting and leaving other sectors at 5% automation. “Full software” starts with software automation already at 100%, and increases automation in every other sector from 0 to  $f$ .

The singularity date goes to the infinite future exactly at the automation thresholds for explosive growth implied by (60) (and as shown in table 4 for the first two experiments): 13% when all sectors are automated, 17% when only software and hardware sectors are automated, 67% when a software automation shock occurs while other sectors remain at 5% automation, and 0% when software is fully automated and the shock size varies across other sectors. This figure also illustrates that under a large automation shock, the growth model predicts a singularity very quickly. It is in these very high-growth scenarios that bottlenecks – ignored in this section – would become a particularly significant drag on growth. We turn to analyzing the role of bottlenecks in the following section.

## 5.4 Limitations of calibration and simulation

There are a number of limitations to the estimates used to calibrate our model.

First, the degree of diminishing returns in software from Erdil, Besiroglu, and Ho (2024) is estimated in software domains other than frontier AI research. Specifically, these estimates come from chess engines and computer vision. Estimating  $r_S$  directly from the

rate of progress at frontier AI labs is difficult, in part because of challenges associated with measuring research inputs. [Ho and Whitfill \(2025\)](#) attempt to make this calculation directly, finding  $r_S$  in the range of 1.2-1.8. Given the limitations of this approach, we rely on the relatively conservative estimate of  $r_S = 1$  from [Erdil, Besiroglu, and Ho \(2024\)](#).

Second, we assume that these parameters are foundational to the knowledge discovery process, rather than the *human* knowledge discovery process. For example, we assume that diminishing returns to research ( $\beta$ ) and parallelization of research ( $\lambda$ ) are independent of whether it is humans or AI completing that scientific research. [Trammell and Korinek \(2026\)](#) suggest reasons why both of these parameters may be different under AI R&D. For example, as identified by [Ekerdt and Wu \(2025\)](#), increasing the researcher share of population may result in an effective strengthening of the ideas-getting-harder-to-find effect as the average quality of researchers declines; in the case of AI we might expect constant effective researcher quality, increasing  $r$  for AI researchers relative to human researchers.

Third, we take estimates from [Bloom et al. \(2020\)](#) and [Erdil, Besiroglu, and Ho \(2024\)](#) as given. Specifically, the estimate of  $\beta_A$ , used to calibrate  $r_A$ , from [Bloom et al. \(2020\)](#) is based on an aggregate estimate of the diminishing returns to research across the whole economy. However, research that has driven advances in hardware and software has presumably also contributed to aggregate TFP. Since we separate out software and hardware progress from TFP, we should also adjust  $r_A$  to be net of software and hardware sectors. We do not have a good estimate of the elasticity of aggregate technological progress that has come from AI software and hardware research ( $\tau_S$  and  $\tau_H$  terms from above), hence we assume these are small and calibrate  $r_A$  using the aggregate economy estimate.<sup>22</sup>

Fourth, our baseline environment (sections 5.2 and 5.3) assumes conventional capital is not a research input, so automation creates *new* feedback loops between output and technology. An alternative specification redefines effective R&D labor as  $\hat{L}_i \propto K_i^{\gamma_i} (L_i^{1-f_i} L_{AI,i}^{f_i})^{1-\gamma_i}$  where  $\gamma_i$  is the elasticity of sector  $i$  research input with respect to conventional capital. Including conventional capital would introduce economic feedback loops that exist even without automation ( $f_i = 0$ ). The techniques of section 3.2 could be used to derive explosive and balanced growth conditions as a function of the research capital shares  $\gamma_i$ .

Fifth, the simulation holds the share of total compute devoted to hardware and software research fixed at recent levels. [Erdil et al. \(2025\)](#) endogenize these shares.

---

<sup>22</sup>A back-of-the-envelope correction is the following. The  $\beta_A = 3.1$  calibration of [Bloom et al. \(2020\)](#) comes from a balanced growth path calibration, where  $\beta_A = n_A/g_A = 4\%/1.3\%$  in their sample. Roughly half of TFP growth since 1989 has come from information technology sectors: that is, TFP growth in non-IT sectors was  $g_A \approx 0.65\%$ . As an upper bound, then, the non-IT part of the innovation network could be calibrated to have  $\tilde{\beta}_A = 4\%/0.65\% = 6.2$ . Recall that under our baseline calibration, 13% automation is required across all sectors to achieve hyperbolic growth. Using a recalibrated version of (60) with  $\tilde{\beta}_A = 6.2$ , hyperbolic growth would require 14% automation across sectors.

Sixth, the model here is formulated in continuous time and abstracts from lags in feedback loops. [Ord \(2026\)](#) highlights that in a discrete time environment – where feedback loops occur through discrete cycles – a singularity requires that the duration of those cycles fall sufficiently quickly toward zero. A positive lower bound on the length of feedback cycles (e.g., due to the amount of time to build new semiconductor fabrication plants) can rule out a literal mathematical singularity, even while allowing superexponential growth.

Finally, these simulations assume an elasticity of substitution of one between human and AI labor, ruling out bottlenecks. We turn to studying this issue in the next section.

## 6 The role of bottlenecks

This section discusses (i) how to think about bottlenecks conceptually, (ii) the key economic object which determines whether a growth explosion is possible, and (iii) a condition on the speed of automation under which the above explosion results continue to hold in the presence of bottlenecks.

The model in section 5 studies an economy without any bottlenecks. Formally, this is evident in (50), where effective labor is a Cobb-Douglas aggregate of human labor and AI labor, with  $\hat{L} \propto L^{1-f} L_{AI}^f$ . That is, the elasticity of substitution between human labor and AI labor is one, so that an infinite amount of either input is sufficient for infinite output. An elasticity less than one would imply that output is bottlenecked by the slower-growing input – in particular, no amount of AI labor can compensate for a shortage of human labor.

Thus, as is well-known, a bottlenecked production technology will never allow for an economic singularity except under full automation. In particular, there are no values of the AI-related diminishing return parameters  $\beta_S$  and  $\beta_H$  in the idea production functions (53)-(54) which would allow for a singularity. Even increasing returns,  $\beta_i < 0$ , would be insufficient for an economic singularity. That’s because, as the level of AI labor  $L_{AI}$  skyrockets to infinity, the level of effective labor  $\hat{L}$  is still bottlenecked by the amount of human labor. In the limit,  $\hat{L} \propto L$ .

Of course, even if the elasticity of substitution were one “in reality” – so that singularities were possible as in the model – then reality could not feature a *true* singularity: to the best of our understanding, we exist in a finite light cone ([Sandberg and Manheim, 2021](#)). Thus, in *any* case, the model here should be understood like any other economic model: an approximation to reality, with a bounded domain of applicability, aiming to serve as a powerful intuition pump ([Krugman, 1998](#)). In *any* case, the conditions derived here could only possibly delineate a temporary period of ‘explosive’ growth – even if the ‘temporary’ period were a very long one.

**Introducing bottlenecks.** It is nonetheless interesting to think about how the presence of explicit bottlenecks affects our application in section 5. Allowing for bottlenecks would amount to replacing the aggregator for effective labor (50) with the following CES aggregator:

$$\hat{L}_i = \left( (1 - f_i)^{\frac{1}{\theta}} L_i^{\frac{\theta-1}{\theta}} + f_i^{\frac{1}{\theta}} L_{AI,i}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}} p_i \quad (62)$$

with  $\theta < 1$  allowing for bottlenecks. Meanwhile,  $\theta \rightarrow 1$  recovers our baseline specification.

The key insight is that the important object is the *elasticity of effective labor to AI*:

$$\eta_i \equiv \frac{\partial \log \hat{L}_i}{\partial \log L_{AI,i}} \quad (63)$$

In the Cobb-Douglas case, this elasticity is simply the degree of automation,  $\eta_i = f_i$ .

In the CES case, this elasticity becomes more complicated. The elasticity of effective labor to AI in the presence of bottlenecks is:

$$\eta_i = \left[ 1 + \left( \frac{1 - f_i}{f_i} \right)^{\frac{1}{\theta}} \left( \frac{L_{AI,i}}{L_i} \right)^{\frac{1-\theta}{\theta}} \right]^{-1} \quad (64)$$

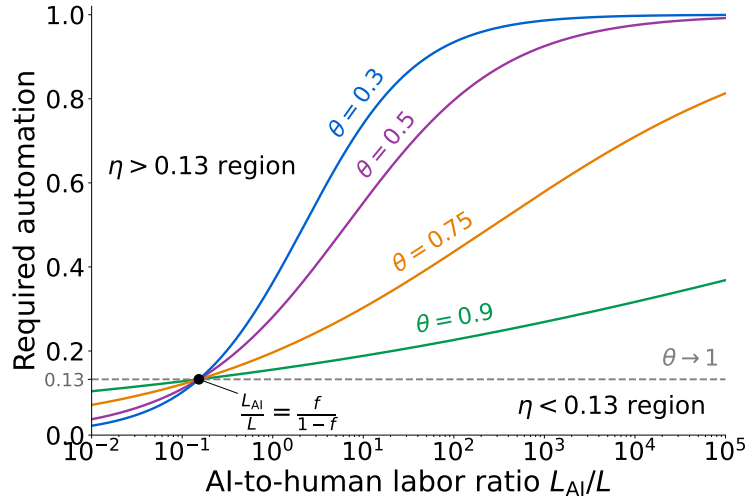
This elasticity is now sensitive not only to the share of automated tasks  $f_i$ , but also it is negatively related to the AI-human labor ratio,  $L_{AI,i}/L_i$ . As a result, as AI labor grows faster than does human labor, this elasticity declines – and so output becomes less and less responsive to increases in the AI labor supply.

In the limit as  $L_{AI,i}/L_i \rightarrow \infty$ , the elasticity  $\eta_i \rightarrow 0$ . In figure 1 of the intro, this would be visually depicted as the arrow from “AI labor” to “effective labor” disappearing, thus severing the AI side of the economy from the traditional portion of the economy and final output. The feedback loop between AI and economic output would be broken.

In the bottlenecked economy, the explosion condition can be satisfied by replacing the benchmark automation terms  $f_i$  with the corresponding CES elasticities of (64),  $\eta_i$ , and requiring that the condition holds at all points in time. The elasticity is a function of not just the level of automation  $f_i$  but also the AI-to-human labor ratio  $L_{AI,i}/L_i$ . As  $L_{AI,i}/L_i$  rises due to software  $S$  improving or compute  $C$  accumulating, the elasticity declines. Thus growth in effective labor tapers off, preventing a growth explosion. However, if automation  $f_i$  is continuously advancing, then this can be an offsetting force and prevent the elasticity from collapsing.

In section 5 we demonstrated that, without bottlenecks and under uniform automa-

Figure 9: The amount of automation required to reach the critical elasticity ( $\eta = 0.13$ ) as a function of the AI-to-human labor ratio,  $L_{AI}/L$ , given elasticity of substitution  $\theta$ .



tion, 13% of every sector of the economy needs to be automated ( $f_i \geq 13\%$ ) for explosive growth. In the presence of bottlenecks, the analogous threshold applies to the elasticity ( $\eta_{i,t} \geq 0.13$  for all  $t$ ). Figure 9 plots the level of automation required to meet the critical threshold  $\eta = 0.13$  at different AI-to-human labor ratios. The region above each colored line exceeds the critical elasticity while the region below fails to meet it. As AI labor becomes more abundant relative to humans, more substantial automation is required to preserve the critical elasticity, and the required increase is steeper the stronger the complementarity (lower  $\theta$ ). With weaker bottlenecks ( $\theta$  close to one) much lower automation suffices, and as  $\theta \rightarrow 1$  the elasticity is sustained at any labor ratio so long as  $f \geq 13\%$ . All frontiers intersect at the point where  $L_{AI}/L = f/(1 - f)$ ; this is the point where the densities of AI and human workers are equal, so there is no bottleneck effect.

**An approximate, local calibration.** The elasticity  $\eta_i$  is empirically measurable using cost shares, as emphasized by Cunningham et al. (2026). Denote spending on human labor plus spending on inference compute as the total budget for effective labor. The fraction of this total budget going to inference compute exactly equals the elasticity  $\eta_i$ , under cost minimization and given constant returns to scale in  $\hat{L}_i$ . Importantly, this is a local measure, identifying the elasticity at the current AI-to-human labor ratio.

OpenAI (2026) reported that, as of August 2026, the median researcher at OpenAI was using over \$600/day worth of AI agent inference compute, or approximately \$200k/year. The Levels.fyi platform reports that the median research scientist at OpenAI earns ap-

proximately \$800k/year in total compensation (Levels.fyi, 2026).<sup>23</sup> Taking these figures as representative implies a local elasticity in the software research sector (to which OpenAI belongs) of approximately  $\eta_S = 20\%$ . However, the August 2026 OpenAI data also shows inference spending per researcher, starting from near zero in February 2026, doubling approximately every six weeks. Reasonably assuming that researcher compensation rose much more slowly than this, the implied estimate of  $\eta_S$  has risen quickly.<sup>24</sup>

**A sufficient condition for preserving the explosion threshold.** We now introduce a sufficient condition under which our baseline results go through even under  $\theta < 1$ . The key change is that, rather than treating the level of automation  $f$  as fixed, we suppose it evolves dynamically. Continual advancement in  $f$  then prevents a decline in the key elasticity  $\eta$ .

A condition on *the speed of automation* that is sufficient to prevent the elasticity from collapsing is the following:

$$\frac{g_{f,i}}{1 - f_i} \geq (1 - \theta)g_{L_{AI,i}/L_i} \quad (65)$$

The left-hand side is the automation speed.<sup>25</sup> Faster automation is required when complementarity is stronger ( $\theta$  smaller) or when the AI-to-human labor ratio is growing faster.

As long as automation progresses at a speed no less than this threshold, the elasticity (64) will not fall. Thus, if the relevant elasticities satisfy the critical threshold at some date, and if condition (65) holds thereafter, then the bottlenecked economy remains on the explosive side of the benchmark threshold. Proposition 3 states these arguments formally.

**Proposition 3 (Bottlenecks).** *Consider the bottlenecked AI-economic model described by equations (49), (62), and (51)-(58). Suppose the sectoral labor elasticities,  $\eta_Y, \eta_S, \eta_H, \eta_A$ , at some time  $T$  satisfy the non-bottlenecked threshold condition*

$$\eta_{Y,T} + \eta_{S,T} \cdot r_S + \eta_{H,T} \cdot r_H + \frac{1}{1 - \alpha} \eta_{A,T} \cdot r_A > 1 \quad (66)$$

*Assume depreciation rates satisfy  $\delta_K^\alpha \delta_C^{1-\alpha} \leq A_T a_K^\alpha \left( a_C H_T p_Y \frac{C_{Y,T}}{C_T} S_T \right)^{1-\alpha}$ . If the speed of automation in each sector satisfies (65) for all  $t \geq T$ , then the system exhibits hyperbolic growth.*

*Proof.* See Appendix B.3. □

<sup>23</sup>Levels.fyi collects user submissions of compensation levels, verified with tax forms or other measures.

<sup>24</sup>OpenAI (2026) reports expenditure using API prices, which include markups, while  $\eta_S$  should be calibrated using cost shares, which exclude markups. Private-sector estimates of OpenAI inference margins range from 0% to 80%, implying estimates of  $\eta_S$  that range from approximately 5% to 20%.

<sup>25</sup>The left-hand side is the speed at which the non-automated/automated task ratio is shrinking.

Condition (66) is simply our prior threshold condition (59), but replacing the  $f_i$  automation terms with the more general  $\eta_{i,t}$  elasticities, and condition (65) guarantees the elasticities (64) are non-decreasing over time. The depreciation restriction is a technical condition to ensure that compute and capital stocks do not go to zero. Then, effective labor under the CES labor aggregator (62) admits a uniform lower bound of Cobb-Douglas form with fixed exponents  $\eta_{i,T}$ . The CES economy then dominates a Cobb-Douglas comparison system with automation shares  $f_i$  set equal to  $\eta_{i,T}$ . Since that comparison system satisfies the threshold in Corollary 3, it explodes in finite time, and so does the CES economy.

**Quantitative implications of bottlenecks.** We can quantitatively simulate the automation speed necessary to keep each sector's elasticity from falling, as in (65). Different sectors have different required automation speeds because the human labor force grows at different speeds across sectors in the simulation ( $n_i$ ); with faster human labor growth, humans bottleneck progress less.

Consider an economy starting with (say) 15% automation in every sector. This is above the 13% threshold for explosive growth in a Cobb-Douglas economy ( $\theta = 1$ ), and – as demonstrated quantitatively in figure 7 – in such an economy without bottlenecks there would be a singularity in 16 years. In an economy with bottlenecks ( $\theta < 1$ ), a fixed level of automation would result in growth petering out.

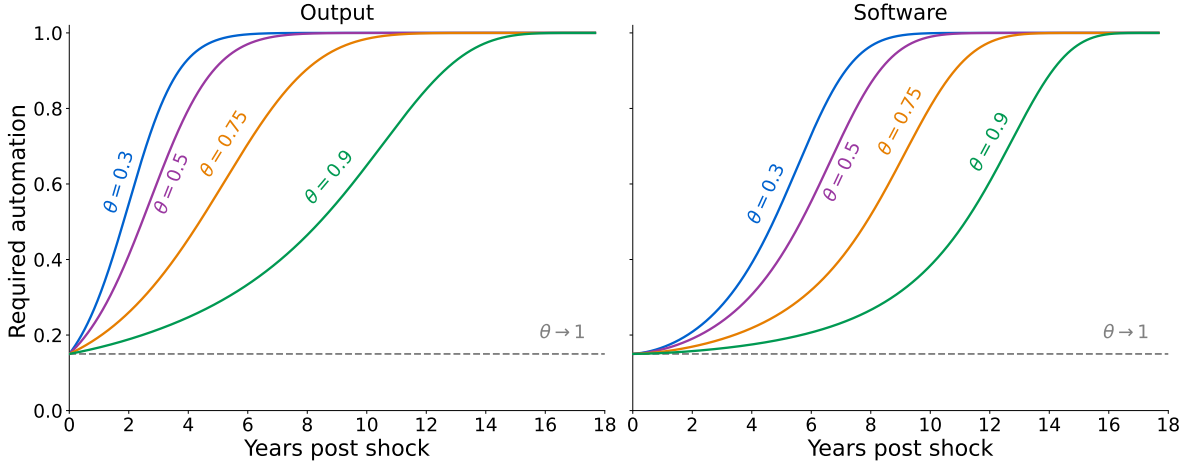
Figure 10 shows how quickly automation must progress in the output sector  $f_{Y,t}$  and in the software sector  $f_{S,t}$  to sustain explosive growth (the hardware and general ideas sectors are omitted).<sup>26</sup> One observes that automation needs to advance more slowly in software than in goods production, because we had assumed that – in line with historical trends – the human labor force in the software sector grows much faster than in goods production ( $n_S = 110\%$  while  $n_Y = 1\%$ ). The rapid hiring of humans in the software sector helps to prevent humans from becoming as severe a bottleneck. We can also observe that, naturally, faster automation is required when the elasticity of substitution is lower.

Two remarks are worth noting. First, if  $L_{AI,i}/L_i \rightarrow \infty$ , then under CES aggregation  $\eta_i \rightarrow 0$  unless  $f_i \rightarrow 1$ , so automation approaching 100% is required to keep AI's output elasticity bounded away from zero. This is evident in figure 10. Second, if some tasks are never automatable, then the elasticity condition will eventually fail, even if the CES economy initially shadows the explosive benchmark.

---

<sup>26</sup>Here, for illustration we normalize the initial level of AI labor such that  $L_{AI,i,0}/L_{i,0} = f_{i,0}/(1 - f_{i,0})$ , so that (as indicated in figure 9)  $\eta_{i,0} = f_{i,0}$  in all sectors, independently of the choice of  $\theta$ .

Figure 10: Minimum automation necessary to sustain an explosive path at  $\eta_i = 0.15$



**Summary.** Together, these points indicate that the possibility of bottlenecks does not invalidate the condition for explosive growth derived above. Our main results consider the Cobb-Douglas case because it has the straightforward property that the elasticity of the aggregator of AI-equivalent labor and human labor is just the share of tasks that can be automated. When human and AI labor are complements, the interpretation of the explosion threshold with automation instead simply uses a different primitive: explosive growth requires a sufficient *weight* on AI in its deployment across research and production.

## 7 Conclusion

“Advanced AI is interesting for many reasons, but perhaps nothing is quite as significant as the fact that we can use it to do faster AI research.” (Altman, 2025)

This paper develops a model for understanding how advances in artificial intelligence could transform economic growth. By considering the interconnected roles of hardware, software, and general technological progress, we show that AI-driven automation could generate feedback effects leading to explosive growth. Using historical estimates of diminishing returns, our model suggests that these effects could be substantial, particularly from the automation of computing hardware R&D. These findings have several implications.

First, our results highlight the strategic importance of semiconductor research and development. The high historical productivity of hardware research, combined with growing automation capabilities in chip design, suggests that this sector could play a central role in determining the pace of overall technological progress. This raises important ques-

tions about both the market concentration and geographic concentration of semiconductor research and production capabilities ([Korinek and Vipra, 2025](#)).

Second, our analysis suggests that monitoring the level of automation in AI R&D is very important and could serve as an early signal of a potential growth acceleration. This is something AI companies could measure and share publicly ([Cunningham et al., 2026](#)).

There is therefore significant scope for further empirical work. These models ultimately require very few parameters, with just a handful pinning down both the explosive growth conditions and the balanced growth paths. Most important is the degree of diminishing returns to research, for which we rely on fairly limited evidence.

## References

- Acemoglu, Daron, and Pascual Restrepo.** 2018. "The race between man and machine: Implications of technology for growth, factor shares, and employment." *American economic review*, 108(6): 1488–1542.
- Acemoglu, Daron, and Pascual Restrepo.** 2019. "Artificial intelligence, automation, and work." In *The Economics of Artificial Intelligence: An Agenda*, ed. Ajay Agrawal, Joshua Gans, and Avi Goldfarb, 197–236. University of Chicago Press.
- Aghion, Philippe, Benjamin F. Jones, and Charles I. Jones.** 2019. "Artificial Intelligence and Economic Growth." In *The Economics of Artificial Intelligence: An Agenda*, ed. Ajay Agrawal, Joshua Gans, and Avi Goldfarb, 237–282. University of Chicago Press.
- Altman, Sam.** 2025. "The Gentle Singularity." *Blog post*.
- Besiroglu, Tamay, Nicholas Emery-Xu, and Neil Thompson.** 2024. "Economic impacts of AI-augmented R&D." *Research Policy*, 53(7): 105037.
- Bloom, Nicholas, Charles I. Jones, John Van Reenen, and Michael Webb.** 2020. "Are ideas getting harder to find?" *American Economic Review*, 110(4): 1104–1144.
- Chen, Xugan, Ernest Liu, and Song Ma.** 2026. "The Economic Value of Science."
- Cunningham, Tom, Lukas Althoff, Basil Halperin, Brian Jabarian, Andrew Koh, Arjun Ramani, Phil Trammell, Parker Whitfill, and Cheryl Wu.** 2026. "The Economics of Recursive Self-Improvement."
- Davidson, Tom.** 2023. "What a Compute-Centric Framework Says About Takeoff Speeds." Open Philanthropy.
- Donald, Eric.** 2023. "Spillovers and the direction of innovation: An Application to the Clean Energy Transition."
- Ekerdt, Lorenz, and Kai-Jie Wu.** 2025. "Self-Selection and the Diminishing Returns of Research."
- Epoch AI.** 2026. "Trends in Artificial Intelligence." Updated February 5, 2026. Accessed March 29, 2026.
- Erdil, Ege, Andrei Potlogea, Tamay Besiroglu, Edu Roldan, Anson Ho, Jaime Sevilla, Matthew Barnett, Matej Vrzla, and Robert Sandler.** 2025. "GATE: An Integrated Assessment Model for AI Automation."

- Erdil, Ege, Tamay Besiroglu, and Anson Ho.** 2024. "Estimating Idea Production: A Methodological Survey." *arXiv preprint arXiv:2405.10494*.
- Goldie, Anna, and Azalia Mirhoseini.** 2024. "How AlphaChip Transformed Computer Chip Design." *Google DeepMind Blog*.
- Good, Irving John.** 1966. "Speculations Concerning the First Ultraintelligent Machine." In *Advances in Computers*, Vol. 6, ed. Franz L. Alt and Morris Rubinoff, 31–88. Elsevier.
- Growiec, Jakub.** 2007. "Beyond the linearity critique: the knife-edge assumption of steady-state growth." *Economic Theory*, 31(3): 489–499.
- Hanson, Robin.** 2000. "Long-term growth as a sequence of exponential modes." *Working manuscript*.
- Ho, Anson, and Parker Whitfill.** 2025. "The software intelligence explosion debate needs experiments." *Epoch AI Substack (Gradient Updates)*.
- Ho, Anson, Tamay Besiroglu, Ege Erdil, David Owen, Robi Rahman, Zifan C. Guo, David Atkinson, Neil Thompson, and Jaime Sevilla.** 2024. "Algorithmic progress in language models." *Advances in Neural Information Processing Systems*, 37: 58245–58283.
- Jones, Ben.** 2025. "Artificial Intelligence in Research and Development." National Bureau of Economic Research.
- Jones, Charles I.** 1995. "R&D-based models of economic growth." *Journal of Political Economy*, 103(4): 759–784.
- Jones, Charles I.** 2026. "A.I., Automation, and Explosive Growth: A Condition for Finite-Time Blow-Up."
- Jones, Charles I., and Christopher Tonetti.** 2025. "Past Automation and Future A.I.: How Weak Links Tame the Growth Explosion." Working paper, Stanford GSB and NBER.
- Korinek, Anton, and Jai Vipra.** 2025. "Concentrating intelligence: scaling and market structure in artificial intelligence." *Economic Policy*, 40(121): 225–256.
- Krugman, Paul.** 1998. "Two cheers for formalism." *The Economic Journal*, 108(451): 1829–1836.
- Levels.fyi.** 2026. "OpenAI Research Scientist Salary." Accessed September 6, 2026.
- Liu, Ernest, and Song Ma.** 2024. "Innovation networks and R&D allocation." National Bureau of Economic Research.
- Mirhoseini, Azalia, Anna Goldie, Mustafa Yazgan, Joe Wenjie Jiang, Ebrahim Songhori, Shen Wang, Young-Joon Lee, Eric Johnson, Omkar Pathak, Azade Nazi, et al.** 2021. "A graph placement methodology for fast chip design." *Nature*, 594(7862): 207–212.
- Ngai, L. Rachel, and Roberto M. Samaniego.** 2011. "Accounting for research and productivity growth across industries." *Review of Economic Dynamics*, 14(3): 475–495.
- OpenAI.** 2025. "Sam, Jakub, and Wojciech on the future of OpenAI with audience Q&A." *YouTube livestream*, Featuring Sam Altman (CEO), Jakub Pachocki (Chief Scientist), and Wojciech Zaremba.
- OpenAI.** 2026. "Research Acceleration: The View Inside OpenAI."
- Ord, Toby.** 2026. "The Dynamics of Intelligence Explosions." *arXiv preprint arXiv:2608.14426*.
- Roodman, David.** 2020. "On the probability distribution of long-term changes in the growth rate of the global economy: An outside view." *Open Philanthropy Report*.
- Sandberg, Anders, and David Manheim.** 2021. "What Is the Upper Limit of Value?"

- Trammell, Philip, and Anton Korinek.** 2026. “Economic Growth Under Transformative AI.” *Annual Review of Economics*, 18: 589–611.
- Yudkowsky, Eliezer.** 2013. “Intelligence explosion microeconomics.” *Machine Intelligence Research Institute Technical Report*, 1.
- Zeira, Joseph.** 1998. “Workers, machines, and economic growth.” *The Quarterly Journal of Economics*, 113(4): 1091–1117.

## A Proof of Proposition 1

*Proof of Proposition 1.* First, suppose  $\mathbf{F}^A$  is irreducible, and let  $\mathbf{u} \gg 0$  be its Perron-Frobenius right eigenvector,  $\mathbf{F}^A \mathbf{u} = \rho(\mathbf{F}^A) \mathbf{u}$ , and set  $\mu \equiv \rho(\mathbf{F}^A) - 1$  and  $m = \min_i \beta_i u_i > 0$ .

Divide equation (20) by  $A_i$  to get technology growth rates and then take logs and time derivatives to get the rate of change in growth rates given by the vector

$$\dot{\mathbf{g}}_A = \text{diag}(\mathbf{g}_A)[(\mathbf{S} - \mathbf{B})\mathbf{g}_A + \text{diag}(\boldsymbol{\lambda})\mathbf{g}_R], \quad (67)$$

where  $S_{i,j} = \phi_{i,j}$  for  $i \neq j$  and zero for diagonal elements, and  $\mathbf{B} = \text{diag}(\beta_1, \beta_2, \dots, \beta_N)$ . We define the ‘exponent matrix’,  $\boldsymbol{\Omega} \equiv \mathbf{S} - \mathbf{B}$ . Therefore, we can relate the technological feedback matrix  $\mathbf{F}^A$  and the exponent matrix  $\boldsymbol{\Omega}$  according to  $\boldsymbol{\Omega} = \mathbf{B}(\mathbf{F}^A - \mathbf{I})$ . Note that  $\boldsymbol{\Omega} \mathbf{u} = \mathbf{B}(\mathbf{F}^A - \mathbf{I}) \mathbf{u} = \mu \mathbf{B} \mathbf{u}$ . For notational clarity, we switch to using parentheticals to denote time:  $z_i(t)$  is variable  $z_i$  at time  $t$ .

*Proving  $\rho(\mathbf{F}^A) > 1 \implies$  hyperbolic growth.* Suppose  $\rho(\mathbf{F}^A) > 1$ , so  $\mu > 0$ , and define  $c_1 = \frac{\mu m}{2} > 0$ . Consider the scalar differential equation that diverges in finite time

$$\dot{z} = c_1 z^2, \quad \text{setting } z(0) = \min_i \frac{g_{A_i}(0)}{u_i} > 0.$$

We claim that  $g_{A_i}(t) \geq z(t)u_i$  for all  $i$ , for as long as the original solution exists. This holds initially by construction. Suppose instead that some component  $i$  were the first to cross below  $z(t)u_i$ . At the first crossing,  $g_{A_i} = zu_i$  and  $g_{A_j} \geq zu_j$  for all  $j$ . Since the off-diagonal elements of  $\boldsymbol{\Omega}$  are non-negative,  $(\boldsymbol{\Omega} \mathbf{g}_A)_i \geq z(\boldsymbol{\Omega} \mathbf{u})_i = z\mu\beta_i u_i$ . Hence, using (67) and the preceding,

$$\begin{aligned} \dot{g}_{A_i} &\geq g_{A_i}(\boldsymbol{\Omega} \mathbf{g}_A)_i \geq z^2 \mu \beta_i u_i^2 \\ &> c_1 z^2 u_i = \dot{z} u_i, \end{aligned}$$

where the strict inequality follows from  $c_1 < \mu\beta_i u_i$ . Thus  $g_{A_i}$  is growing faster than  $zu_i$  at any potential first crossing, so such a crossing cannot occur. Therefore,  $\mathbf{g}_A(t) \geq z(t)\mathbf{u}$ , and

thus, since  $z(t)$  explodes in finite time, so too does  $\mathbf{g}_A$ .

Moreover, at least one technology level diverges in finite time. If all technology levels remained bounded, growth rates would also remain bounded by (20).

*Proving  $\rho(\mathbf{F}^A) \leq 1 \implies$  no hyperbolic growth.* Continue to suppose that  $\mathbf{F}^A$  is irreducible, but now suppose that  $\rho(\mathbf{F}^A) \leq 1$ , so  $\mu \leq 0$ . Choose some constant  $c_2 \geq \max_i \{\lambda_i g_{R_i}\}$ . Let  $z$  solve

$$\dot{z} = \mu m z^2 + c_2 z, \quad z(0) = \max_i \frac{g_{A_i}(0)}{u_i}.$$

We claim that  $g_{A_i}(t) \leq z(t)u_i$  for all  $i$ . Again, this holds initially. At any potential first crossing from below,  $g_{A_i} = zu_i$ , and  $g_{A_j} \leq zu_j$  for all  $j$ , and hence  $(\Omega \mathbf{g}_A)_i \leq z(\Omega \mathbf{u})_i = z\mu\beta_i u_i$ . Therefore,

$$\begin{aligned} \frac{\dot{g}_{A_i}}{u_i} &\leq \mu\beta_i u_i z^2 + \lambda_i g_{R_i} z \leq \mu m z^2 + \lambda_i g_{R_i} z \\ &\leq \mu m z^2 + c_2 z = \dot{z}. \end{aligned}$$

Thus no component can cross above  $zu_i$ .

If  $\mu < 0$ ,  $z$  follows a logistic differential equation with a negative quadratic term and thus remains bounded for all  $t$ . Hence technology growth rates remain bounded and there is no finite-time explosion. If  $\mu = 0$ , then  $z$  grows exponentially, so technology growth rates grow at most exponentially and technology levels at most double exponentially.

*Balanced growth when  $\rho(\mathbf{F}^A) < 1$  and not when  $\rho(\mathbf{F}^A) = 1$ .* When  $\rho(\mathbf{F}^A) < 1$ , the BGP growth rates derived in the main text are strictly positive. Substituting the corresponding exponential paths into (20) gives a linear system in  $\log \mathbf{A}(0)$  with invertible coefficient matrix  $\Omega$ , establishing existence.

When  $\rho(\mathbf{F}^A) = 1$ , let  $\mathbf{w} \geq 0$  be a nonzero left Perron-Frobenius eigenvector, so  $\mathbf{w}'\mathbf{F}^A = \mathbf{w}'$ . The necessary BGP condition would imply  $0 = \mathbf{w}'(\mathbf{I} - \mathbf{F}^A)\mathbf{g}_A^{\text{BGP}} = \mathbf{w}'\mathbf{r}\mathbf{g}_R > 0$ , a contradiction. Hence no balanced growth path exists.

*Generalizing to the reducible case.* So far, we have assumed  $\mathbf{F}^A$  is irreducible. Now allow  $\mathbf{F}^A$  to be reducible. Both the balanced-growth arguments also apply to reducible networks, but the explosive case needs to be generalized.

Partition the sectors into strongly connected components  $I_1, \dots, I_M$ , ordered so that  $\mathbf{F}^A$  is block lower triangular. Each diagonal block  $\mathbf{F}_{kk}^A$  is irreducible, and  $\rho(\mathbf{F}^A) = \max_{k=1, \dots, M} \rho(\mathbf{F}_{kk}^A)$ .

If  $\rho(\mathbf{F}^A) > 1$ , there is a component  $I_k$  with  $\rho(\mathbf{F}_{kk}^A) > 1$ . Apply the preceding lower-bound argument to this component, using its strictly positive Perron-Frobenius eigenvector. Spillovers from other components are nonnegative, so they only strengthen the lower bound. Hence at least one technology diverges in finite time.

Suppose instead that  $\rho(\mathbf{F}^A) \leq 1$ . We prove non-explosion component by component, in the ordering above. Write  $\mathbf{g}_{A,k}$  for the vector of growth rates in  $I_k$  and  $\mathbf{B}_k = \text{diag}(\beta_i : i \in I_k)$ . The dynamics within this component are  $\dot{\mathbf{g}}_{A,k} = \text{diag}(\mathbf{g}_{A,k}) [\mathbf{B}_k(\mathbf{F}_{kk}^A - \mathbf{I})\mathbf{g}_{A,k} + \mathbf{d}_k(t)]$ , where, for  $i \in I_k$ ,  $d_{k,i}(t) = \lambda_i g_{R_i} + \sum_{\ell < k} \sum_{j \in I_\ell} \phi_{i,j} g_{A_j}(t)$ . For the first component, the sum is empty. For each subsequent component, suppose inductively that all preceding components are non-explosive. On any finite interval  $[0, T]$ , their growth rates are bounded, so there is a finite constant  $C_{k,T} \geq 0$  such that  $d_{k,i}(t) \leq C_{k,T}$  for all  $i \in I_k$  and  $t \in [0, T]$ .

Let  $\mathbf{u}^{(k)} \gg 0$  be a Perron–Frobenius right eigenvector of  $\mathbf{F}_{kk}^A$ . Since  $\rho(\mathbf{F}_{kk}^A) \leq 1$ ,  $\mathbf{B}_k(\mathbf{F}_{kk}^A - \mathbf{I})\mathbf{u}^{(k)} = [\rho(\mathbf{F}_{kk}^A) - 1]\mathbf{B}_k\mathbf{u}^{(k)} \leq 0$ . Consequently, the preceding upper-bound comparison applies with  $\dot{z} = C_{k,T}z$  and  $z(0) = \max_{i \in I_k} g_{A_i}(0)/u_i^{(k)}$ . It gives  $\mathbf{g}_{A,k}(t) \leq z(t)\mathbf{u}^{(k)}$  on  $[0, T]$ . Since  $z(t) = z(0)e^{C_{k,T}t}$  remains finite and  $T$  was arbitrary, this component cannot explode in finite time. The entire network, then, is non-explosive, by induction.

Thus, also when  $\mathbf{F}^A$  is reducible, the system exhibits finite-time explosion if and only if  $\rho(\mathbf{F}^A) > 1$ .

□

# Online-Only Appendix

## B Additional proofs

### B.1 Proof of Proposition 2

*Proof of Proposition 2.* Here we define a new system of differential equations including capital motion. The proof closely follows the same comparison argument as the proof of Proposition 1, applied to an augmented system that includes capital accumulation.

Define the scalar  $x_0 = g_K + \delta$ , the vector  $\mathbf{x} = (x_0 \ \mathbf{g}'_A)'$ , and  $\mathbf{d} = (\delta \ 0 \ \dots \ 0)'$ . Therefore,

$$\mathbf{x} - \mathbf{d} = (g_K \ \mathbf{g}'_A)'$$

Thus, the growth dynamics (explosive or otherwise) of the  $\mathbf{x}$  system are exactly the same as those for the  $(g_K, \mathbf{g}_A)$  system. As in the proof of Proposition 1, we can take logs and time derivatives to obtain

$$\dot{\mathbf{x}} = \text{diag}(\mathbf{x})[\tilde{\Omega}(\mathbf{x} - \mathbf{d}) + \mathbf{c}], \quad (68)$$

where the  $\tilde{\cdot}$  notation is used to differentiate the key matrices from the proof of Proposition 1, and

$$\begin{aligned} \tilde{\Omega} &= \tilde{\mathbf{B}}(\tilde{\mathbf{F}} - \mathbf{I}), & \tilde{\mathbf{B}} &= \text{diag}((1 - \alpha), \beta_1, \dots, \beta_N), \\ \tilde{\mathbf{F}} &= \begin{pmatrix} 0 & \frac{\tau'}{1-\alpha} \\ \mathbf{r}\gamma & \mathbf{F}^A \end{pmatrix}, \end{aligned}$$

and

$$\mathbf{c} = n((1 - \alpha), \lambda_1(1 - \gamma_1), \dots, \lambda_N(1 - \gamma_N))'.$$

First we show that the size of  $\rho(\tilde{\mathbf{F}})$  relative to 1 determines the explosiveness of the system, as did  $\rho(\mathbf{F}^A)$  in the proof of Proposition 1. We then show that the sign of

$$\tilde{\mu} \equiv \rho(\tilde{\mathbf{F}}) - 1$$

is the same as that of  $\rho(\mathbf{F}^A + \mathbf{F}^Y) - 1$ .

In contrast to the proof of Proposition 1, with capital motion we require the additional

result

$$\text{there exists } \bar{t} \text{ such that } g_K(t) > 0 \text{ for all } t > \bar{t}. \quad (69)$$

To see this, suppose first that the solution exists globally. Since labor and technologies are non-decreasing, there exists a constant  $c_K > 0$  such that

$$\dot{K} \geq c_K K^\alpha - \delta K. \quad (70)$$

Therefore,

$$g_K \geq c_K K^{\alpha-1} - \delta.$$

For  $\delta > 0$ , this implies that if

$$K < \left( \frac{\delta}{c_K} \right)^{\frac{1}{\alpha-1}},$$

then  $g_K > 0$ , and hence

$$K(t) \geq \underline{K} \equiv \min \left\{ K(0), \left( \frac{\delta}{c_K} \right)^{\frac{1}{\alpha-1}} \right\} > 0$$

for all  $t$ . If  $\delta = 0$ , then  $g_K > 0$  immediately.

Next, since  $K$ ,  $L$ , and all technologies are bounded away from zero, for each technology there exists some  $\kappa_i > 0$  such that

$$\dot{A}_i(t) \geq \kappa_i A_i(t)^{1-\beta_i}. \quad (71)$$

Therefore,  $A_i(t) \rightarrow \infty$  and hence  $\bar{A} \rightarrow \infty$ . Additionally,

$$g_K = \xi \bar{A} K^{\alpha-1} L^{1-\alpha} - \delta$$

for some constant  $\xi > 0$ . Suppose towards a contradiction that  $g_K(t) \leq 0$  for all  $t$ . Then  $K$  is non-increasing, so  $K^{\alpha-1}$  is non-decreasing, while  $L^{1-\alpha}$  is non-decreasing and  $\bar{A}$  diverges. Therefore

$$g_K + \delta = \xi \bar{A} K^{\alpha-1} L^{1-\alpha} \rightarrow \infty,$$

contradicting  $g_K + \delta \leq \delta$ . Hence  $g_K > 0$  at some finite time.

Finally, since  $x_0 = g_K + \delta$ ,

$$\frac{\dot{x}_0}{x_0} = \boldsymbol{\tau}' \mathbf{g}_A - (1 - \alpha)g_K + (1 - \alpha)n.$$

Thus, whenever  $g_K = 0$ ,

$$\frac{\dot{x}_0}{x_0} = \boldsymbol{\tau}' \mathbf{g}_A + (1 - \alpha)n > 0.$$

Since  $\dot{g}_K = \dot{x}_0$ ,  $g_K$  cannot cross zero from above. Hence there exists some  $\bar{t}$  such that

$$g_K(t) > 0 \quad \text{for all } t > \bar{t}.$$

Therefore, after increasing  $\bar{t}$  if necessary,

$$\mathbf{x}(t) - \mathbf{d} \gg 0 \quad \text{for all } t \geq \bar{t}.$$

*Proving  $\rho(\tilde{\mathbf{F}}) > 1 \implies$  hyperbolic growth.* Suppose  $\tilde{\mathbf{F}}$  is irreducible. Let  $\mathbf{u} \gg 0$  be the Perron–Frobenius right eigenvector of  $\tilde{\mathbf{F}}$ ,

$$\tilde{\mathbf{F}}\mathbf{u} = \rho(\tilde{\mathbf{F}})\mathbf{u}.$$

Then

$$\tilde{\Omega}\mathbf{u} = \tilde{\mathbf{B}}(\tilde{\mathbf{F}} - \mathbf{I})\mathbf{u} = \tilde{\mu}\tilde{\mathbf{B}}\mathbf{u}.$$

Suppose  $\rho(\tilde{\mathbf{F}}) > 1$ , so  $\tilde{\mu} > 0$ , and define

$$m = \min_i \tilde{B}_{ii}u_i > 0, \quad \kappa = \frac{\tilde{\mu}m}{2} > 0.$$

If the original solution already has a finite maximal interval of existence, there is nothing further to prove. Otherwise, the preceding argument gives some  $\bar{t}$  for which  $\mathbf{x}(t) - \mathbf{d} \gg 0$  for all  $t > \bar{t}$ . Consider the scalar differential equation

$$\dot{z} = \kappa z^2, \quad z(\bar{t}) = \min_i \frac{x_i(\bar{t}) - d_i}{u_i} > 0.$$

By the same first-crossing argument as in the proof of Proposition 1,

$$\mathbf{x}(t) - \mathbf{d} \geq z(t)\mathbf{u}$$

for as long as the original solution exists; the argument is unchanged since  $x_i \geq x_i - d_i$  and  $c \geq 0$ . Since  $z$  diverges in finite time, so does at least one component of  $\mathbf{x}$ , and hence at least one of the growth rates  $g_{A_i}$  or  $g_K$ . This requires at least one of  $A_i$  or  $K$  to diverge in finite time.

*Proving  $\rho(\tilde{\mathbf{F}}) \leq 1 \implies$  no hyperbolic growth.* Continue to suppose  $\tilde{\mathbf{F}}$  is irreducible, but now

suppose  $\rho(\tilde{\mathbf{F}}) \leq 1$ , so that  $\tilde{\mu} \leq 0$ . Rewrite the growth-rate system as

$$\dot{\mathbf{x}} = \text{diag}(\mathbf{x})[\tilde{\Omega}\mathbf{x} + \tilde{\mathbf{c}}], \quad \tilde{\mathbf{c}} \equiv \mathbf{c} - \tilde{\Omega}\mathbf{d}.$$

Define

$$m \equiv \min_i \tilde{B}_{ii} u_i > 0$$

and choose a constant

$$C \geq \max \left\{ 0, \max_i \tilde{c}_i \right\}.$$

Let  $z$  solve

$$\dot{z} = \tilde{\mu} m z^2 + C z, \quad z(0) = \max_i \frac{x_i(0)}{u_i}. \quad (72)$$

And again, one can follow the same steps as in the proof for Proposition 1 to show that

$$x_i(t) \leq z(t) u_i \quad \text{for all } i \text{ and } t$$

and the same bounding arguments apply here as in the proof of Proposition 1.

Next we demonstrate that  $\rho(\tilde{\mathbf{F}}) - 1$  has the same sign as  $\rho(\mathbf{F}^A + \mathbf{F}^Y) - 1$ . Write

$$\mathbf{p}' = \frac{\boldsymbol{\tau}'}{1 - \alpha}, \quad \mathbf{q} = \mathbf{r}\boldsymbol{\gamma},$$

so that  $\mathbf{q}\mathbf{p}' = \mathbf{F}^Y$ . Let

$$\lambda = \rho(\tilde{\mathbf{F}}), \quad \mathbf{u} = (u_0 \ \mathbf{u}'_A)'$$

The Perron-Frobenius eigenvalue equation gives

$$\lambda \mathbf{u} = \tilde{\mathbf{F}} \mathbf{u} \quad (73)$$

$$= \begin{pmatrix} \mathbf{p}' \mathbf{u}_A \\ \mathbf{q} u_0 + \mathbf{F}^A \mathbf{u}_A \end{pmatrix}. \quad (74)$$

Solving the top row gives

$$u_0 = \lambda^{-1} \mathbf{p}' \mathbf{u}_A.$$

Therefore the bottom row becomes

$$\left( \mathbf{F}^A + \frac{1}{\lambda} \mathbf{F}^Y \right) \mathbf{u}_A = \lambda \mathbf{u}_A. \quad (75)$$

Since  $\mathbf{u}_A \gg 0$ , Perron–Frobenius implies

$$\lambda = \rho \left( \mathbf{F}^A + \frac{1}{\lambda} \mathbf{F}^Y \right). \quad (76)$$

Since  $\mathbf{F}^Y$  is non-negative,  $\rho(\mathbf{F}^A + \lambda^{-1} \mathbf{F}^Y)$  is weakly decreasing in  $\lambda$ . Therefore,

$$\begin{cases} \rho \left( \mathbf{F}^A + \frac{1}{\lambda} \mathbf{F}^Y \right) \geq \rho(\mathbf{F}^A + \mathbf{F}^Y) & \text{if } \lambda < 1, \\ \rho \left( \mathbf{F}^A + \frac{1}{\lambda} \mathbf{F}^Y \right) = \rho(\mathbf{F}^A + \mathbf{F}^Y) & \text{if } \lambda = 1, \\ \rho \left( \mathbf{F}^A + \frac{1}{\lambda} \mathbf{F}^Y \right) \leq \rho(\mathbf{F}^A + \mathbf{F}^Y) & \text{if } \lambda > 1. \end{cases} \quad (77)$$

This implies that the sign of  $\rho(\tilde{\mathbf{F}}) - 1$  is the same as that of  $\rho(\mathbf{F}^A + \mathbf{F}^Y) - 1$ .

*Balanced growth when  $\rho(\mathbf{F}^A + \mathbf{F}^Y) < 1$  and not when  $\rho(\mathbf{F}^A + \mathbf{F}^Y) = 1$ .* When  $\rho(\mathbf{F}^A + \mathbf{F}^Y) < 1$ , the BGP technology growth rates derived in the main text are strictly positive, since  $\mathbf{n} \gg 0$ . The corresponding capital growth rate is

$$g_K^{\text{BGP}} = n + \frac{\boldsymbol{\tau}' \mathbf{g}_A^{\text{BGP}}}{1 - \alpha} > 0.$$

Substituting the corresponding exponential paths into the technology and capital laws of motion and matching coefficients gives a linear system in  $\log K(0)$  and  $\log \mathbf{A}(0)$  with coefficient matrix  $\tilde{\Omega}$ . This matrix is invertible because

$$\det(\mathbf{I} - \tilde{\mathbf{F}}) = \det(\mathbf{I} - \mathbf{F}^A - \mathbf{F}^Y) \neq 0.$$

Thus suitable positive initial levels exist, establishing existence of a balanced growth path.

When  $\rho(\mathbf{F}^A + \mathbf{F}^Y) = 1$ , let  $\mathbf{w} \geq 0$  be a nonzero left Perron–Frobenius eigenvector, so  $\mathbf{w}'(\mathbf{F}^A + \mathbf{F}^Y) = \mathbf{w}'$ . The necessary BGP condition would imply

$$0 = \mathbf{w}'(\mathbf{I} - \mathbf{F}^A - \mathbf{F}^Y) \mathbf{g}_A^{\text{BGP}} = \mathbf{w}' \mathbf{r} \mathbf{n} > 0,$$

a contradiction. Hence no balanced growth path exists. Both balanced-growth arguments apply to reducible networks.

*Generalizing to the reducible case.* So far, we have assumed that  $\tilde{\mathbf{F}}$  is irreducible. Now allow  $\tilde{\mathbf{F}}$  to be reducible.

Partition the indices  $\{0, \dots, N\}$  into strongly connected components  $I_1, \dots, I_M$ , ordered

so that  $\tilde{\mathbf{F}}$  is block lower triangular. Each diagonal block  $\tilde{\mathbf{F}}_{kk}$  has a strictly positive Perron–Frobenius eigenvector, and

$$\rho(\tilde{\mathbf{F}}) = \max_{k=1,\dots,M} \rho(\tilde{\mathbf{F}}_{kk}).$$

Write  $\tilde{\mathbf{B}}_k = \text{diag}(\tilde{B}_{ii} : i \in I_k)$ , and use subscripts  $k$  to denote the corresponding subvectors.

If  $\rho(\tilde{\mathbf{F}}) > 1$ , there is a component  $I_k$  with  $\rho(\tilde{\mathbf{F}}_{kk}) > 1$ . Suppose towards a contradiction that the solution exists globally. The preceding argument gives a time  $\bar{t}$  such that  $\mathbf{y}(t) \equiv \mathbf{x}(t) - \mathbf{d} \gg 0$  for all  $t \geq \bar{t}$ . Within this component,

$$\dot{\mathbf{y}}_k = \text{diag}(\mathbf{y}_k + \mathbf{d}_k) \left[ \tilde{\mathbf{B}}_k(\tilde{\mathbf{F}}_{kk} - \mathbf{I})\mathbf{y}_k + \mathbf{c}_k + \sum_{\ell < k} \tilde{\mathbf{B}}_k \tilde{\mathbf{F}}_{k\ell} \mathbf{y}_\ell \right].$$

The terms from other components are nonnegative, as is  $\mathbf{c}_k$ . Since  $\mathbf{y}_k + \mathbf{d}_k \geq \mathbf{y}_k$ , the preceding lower-bound argument applies to this component using its strictly positive Perron–Frobenius eigenvector. It yields finite-time explosion, contradicting global existence. Hence at least one technology or capital level diverges in finite time.

Suppose instead that  $\rho(\tilde{\mathbf{F}}) \leq 1$ . We prove non-explosion component by component, in the ordering above. The dynamics within component  $I_k$  are

$$\dot{\mathbf{x}}_k = \text{diag}(\mathbf{x}_k) \left[ \tilde{\mathbf{B}}_k(\tilde{\mathbf{F}}_{kk} - \mathbf{I})\mathbf{x}_k + \mathbf{b}_k(t) \right],$$

where

$$\mathbf{b}_k(t) = \tilde{\mathbf{c}}_k + \sum_{\ell < k} \tilde{\mathbf{B}}_k \tilde{\mathbf{F}}_{k\ell} \mathbf{x}_\ell(t).$$

For the first component, the sum is empty. For each subsequent component, suppose inductively that all preceding components are non-explosive. On any finite interval  $[0, T]$ , their growth rates are bounded, so there is a finite constant  $C_{k,T} \geq 0$  such that  $b_{k,i}(t) \leq C_{k,T}$  for all  $i \in I_k$  and  $t \in [0, T]$ .

Let  $\mathbf{u}^{(k)} \gg 0$  be a Perron–Frobenius right eigenvector of  $\tilde{\mathbf{F}}_{kk}$ . Since  $\rho(\tilde{\mathbf{F}}_{kk}) \leq 1$ ,

$$\tilde{\mathbf{B}}_k(\tilde{\mathbf{F}}_{kk} - \mathbf{I})\mathbf{u}^{(k)} = [\rho(\tilde{\mathbf{F}}_{kk}) - 1]\tilde{\mathbf{B}}_k\mathbf{u}^{(k)} \leq 0.$$

Consequently, the preceding upper-bound comparison applies with

$$\dot{z} = C_{k,T}z, \quad z(0) = \max_{i \in I_k} \frac{x_i(0)}{u_i^{(k)}}.$$

It gives  $\mathbf{x}_k(t) \leq z(t)\mathbf{u}^{(k)}$  on  $[0, T]$ . Since  $z(t) = z(0)e^{C_{k,T}t}$  remains finite and  $T$  was arbitrary, this component cannot explode in finite time. Induction establishes non-explosion for the entire system.

Thus, also when  $\tilde{\mathbf{F}}$  is reducible, the system exhibits finite-time explosion if and only if  $\rho(\tilde{\mathbf{F}}) > 1$ .

The spectral-radius comparison also extends to reducible networks. For  $\varepsilon > 0$ , define

$$\mathbf{p}_\varepsilon = \mathbf{p} + \varepsilon \mathbf{1}, \quad \mathbf{q}_\varepsilon = \mathbf{q} + \varepsilon \mathbf{1}, \quad \mathbf{F}_\varepsilon^A = \mathbf{F}^A + \varepsilon \mathbf{1} \mathbf{1}',$$

and

$$\tilde{\mathbf{F}}_\varepsilon = \begin{pmatrix} 0 & \mathbf{p}'_\varepsilon \\ \mathbf{q}_\varepsilon & \mathbf{F}_\varepsilon^A \end{pmatrix}, \quad \lambda_\varepsilon = \rho(\tilde{\mathbf{F}}_\varepsilon).$$

Since  $\tilde{\mathbf{F}}_\varepsilon$  is irreducible, the preceding spectral-radius argument gives

$$\lambda_\varepsilon = \rho(\mathbf{F}_\varepsilon^A + \lambda_\varepsilon^{-1} \mathbf{q}_\varepsilon \mathbf{p}'_\varepsilon).$$

If  $\lambda = \rho(\tilde{\mathbf{F}}) > 0$ , continuity of the spectral radius gives

$$\lambda = \rho(\mathbf{F}^A + \lambda^{-1} \mathbf{F}^Y),$$

so the same comparison with 1 applies. If  $\lambda = 0$ , then  $\lambda_\varepsilon < 1$  for sufficiently small  $\varepsilon$ , and

$$0 \leq \rho(\mathbf{F}^A + \mathbf{F}^Y) \leq \rho(\mathbf{F}_\varepsilon^A + \mathbf{q}_\varepsilon \mathbf{p}'_\varepsilon) \leq \lambda_\varepsilon \longrightarrow 0.$$

Thus  $\rho(\mathbf{F}^A + \mathbf{F}^Y) = 0$  in this case. Hence  $\rho(\tilde{\mathbf{F}}) - 1$  and  $\rho(\mathbf{F}^A + \mathbf{F}^Y) - 1$  have the same sign, including for reducible networks.

Finally, suppose  $\mathbf{G} \equiv \mathbf{F}^A + \mathbf{F}^Y$  is irreducible and  $\rho(\mathbf{G}) = 1$ . Let  $\mathbf{v} \gg 0$  satisfy  $\mathbf{G}\mathbf{v} = \mathbf{v}$ .

Then

$$\mathbf{u} = (\mathbf{p}'\mathbf{v} \quad \mathbf{v}')' \gg 0, \quad \tilde{\mathbf{F}}\mathbf{u} = \begin{pmatrix} \mathbf{p}'\mathbf{v} \\ \mathbf{q}\mathbf{p}'\mathbf{v} + \mathbf{F}^A\mathbf{v} \end{pmatrix} = \mathbf{u}.$$

Thus the preceding upper-bound argument applies with  $\dot{z} = Cz$ , even if  $\tilde{\mathbf{F}}$  is reducible. Growth rates are bounded by an exponential function, and technology and capital levels by a double-exponential function. □

## B.2 Applying Proposition 2 to Corollaries 1 and 2

Here we present the proof for Corollary 2. Corollary 1 is a special case of Corollary 2 setting  $f_Y = 0$  and  $\tau = 0$ .

*Proof of Corollary 2.* We retain the original positive parameters  $\beta_i$  in  $\mathbf{B} = \text{diag}(\beta_1, \dots, \beta_N)$  and include automation-induced own-sector feedback on the diagonal of the feedback matrix. Define

$$\begin{aligned} \mathbf{q} &\equiv \mathbf{r}\mathbf{f} \\ q_i &\equiv r_i f_i \\ \mathbf{F}_*^A &\equiv \mathbf{q}\boldsymbol{\sigma}' \end{aligned}$$

The growth-rate comparison arguments in Proposition 1 allow nonnegative diagonal entries in the feedback matrix: they require  $\mathbf{B}$  to have positive diagonal entries and the feedback matrix to be nonnegative. Thus they apply with  $\mathbf{F}_*^A$ , including when  $\beta_i - f_i \lambda_i \sigma_i \leq 0$ . Since  $\mathbf{F}_*^A$  has rank at most one,

$$\rho(\mathbf{F}_*^A) = \boldsymbol{\sigma}'\mathbf{q} = \sum_i r_i f_i \sigma_i.$$

This gives the threshold in Corollary 1.

For the economic-feedback calculation, suppose  $f_Y < 1$  and define

$$\begin{aligned} \mathbf{v} &\equiv \frac{1}{1 - f_Y} \left( \frac{\tau}{1 - \alpha} + f_Y \boldsymbol{\sigma} \right), \\ \mathbf{F}_*^Y &\equiv \mathbf{q}\mathbf{v}'. \end{aligned}$$

The combined feedback matrix is

$$\mathbf{G}_* \equiv \mathbf{F}_*^A + \mathbf{F}_*^Y = \frac{\mathbf{q}}{1 - f_Y} \left( \boldsymbol{\sigma} + \frac{\boldsymbol{\tau}}{1 - \alpha} \right)'.$$

It also has rank at most one, so

$$\rho(\mathbf{G}_*) = \frac{1}{1 - f_Y} \sum_i r_i f_i \left( \sigma_i + \frac{\tau_i}{1 - \alpha} \right).$$

Consequently,

$$\rho(\mathbf{G}_*) > 1 \iff f_Y + \sum_i r_i f_i \left( \frac{\tau_i}{1 - \alpha} + \sigma_i \right) > 1.$$

□

### B.3 Proof of Proposition 3

*Proof of Proposition 3.* We first use the depreciation restriction to show that capital and compute remain bounded away from zero. We then construct a Cobb-Douglas comparison economy with fixed automation shares  $\eta_{i,T}$  and apply Corollary 3.

*Bounding capital and compute away from zero.* For  $0 < \theta < 1$ , the CES aggregator (62) satisfies

$$\hat{L}_i \geq p_i \min\{L_i, L_{AI,i}\}.$$

Define the positive constant

$$b \equiv A_T \left( p_Y \frac{C_{Y,T}}{C_T} S_T \right)^{1-\alpha}.$$

The depreciation restriction in the proposition can then be written as

$$\delta_K^\alpha \delta_C^{1-\alpha} \leq a_K^\alpha (a_C H_T)^{1-\alpha} b.$$

Consequently, we can choose positive constants  $\underline{C}$  and  $\underline{K}$  whose ratio satisfies

$$a_K b \left( \frac{\underline{C}}{\underline{K}} \right)^{1-\alpha} \geq \delta_K, \quad a_C H_T b \left( \frac{\underline{C}}{\underline{K}} \right)^{-\alpha} \geq \delta_C.$$

Indeed, when both depreciation rates are positive, the admissible ratios form the nonempty interval

$$\left( \frac{\delta_K}{a_K b} \right)^{\frac{1}{1-\alpha}} \leq \frac{\underline{C}}{\underline{K}} \leq \left( \frac{a_C H_T b}{\delta_C} \right)^{\frac{1}{\alpha}}.$$

Scaling both constants down by the same factor, we can additionally ensure

$$\underline{C} \leq C_T, \quad \underline{K} \leq K_T, \quad \frac{C_{Y,T}}{C_T} \underline{C} S_T \leq L_{Y,T}.$$

The technology stocks  $A$ ,  $S$ , and  $H$  are nondecreasing by their laws of motion, and human labor is nondecreasing because  $n > 0$ . Sectoral compute shares are fixed. Therefore, whenever  $C \geq \underline{C}$  and  $K \geq \underline{K}$ ,

$$Y \geq A_T \left( p_Y \frac{C_{Y,T}}{C_T} \underline{C} S_T \right)^{1-\alpha} \underline{K}^\alpha = b \underline{C}^{1-\alpha} \underline{K}^\alpha.$$

At the boundary  $K = \underline{K}$ , this gives

$$\dot{K} \geq a_K b \underline{C}^{1-\alpha} \underline{K}^\alpha - \delta_K \underline{K} \geq 0.$$

Similarly, at the boundary  $C = \underline{C}$ ,

$$\dot{C} \geq a_C H_T b \underline{C}^{1-\alpha} \underline{K}^\alpha - \delta_C \underline{C} \geq 0.$$

Since both variables start at or above their respective lower bounds, and their rates of change are nonnegative at those bounds, neither variable can cross below its bound. Thus  $C(t) \geq \underline{C}$  and  $K(t) \geq \underline{K}$  for all  $t \geq T$  for which the solution exists. For each sector, define the resulting lower bound on AI labor by

$$\underline{L}_{AI,i} \equiv \frac{C_{i,T}}{C_T} \underline{C} S_T > 0.$$

Then  $L_{AI,i}(t) \geq \underline{L}_{AI,i}$  throughout this interval.

*Constructing a Cobb-Douglas lower bound.* For  $0 < f_i < 1$ , differentiating (64) gives

$$\frac{d}{dt} \log \left( \frac{\eta_i}{1 - \eta_i} \right) = \frac{1}{\theta} \left[ \frac{g_{f,i}}{1 - f_i} - (1 - \theta) g_{L_{AI,i}/L_i} \right] \geq 0,$$

where the inequality follows from (65). Hence  $\eta_i(t) \geq \eta_{i,T}$  for all  $t \geq T$  before explosion.<sup>27</sup>

Combining (62) and (64), we obtain the exact identity, for  $f_i > 0$ ,

$$\hat{L}_i = p_i \eta_i^{\frac{\theta}{1-\theta}} f_i^{-\frac{1}{1-\theta}} L_{AI,i}.$$

If  $\eta_{i,T} > 0$ , then  $f_i \leq 1$ , the elasticity bound, and the lower bound on AI labor imply

$$\begin{aligned} \hat{L}_i(t) &\geq p_i \eta_{i,T}^{\frac{\theta}{1-\theta}} L_{AI,i}(t) \\ &\geq p_i \eta_{i,T}^{\frac{\theta}{1-\theta}} \underline{L}_{AI,i}^{1-\eta_{i,T}} L_{AI,i}(t)^{\eta_{i,T}}. \end{aligned}$$

---

<sup>27</sup>At  $f_i = 0$  or  $f_i = 1$ , interpret (65) in its undivided form,  $\dot{f}_i \geq (1 - \theta) f_i (1 - f_i) g_{L_{AI,i}/L_i}$ . If  $\eta_{i,T} = 0$ , the bound  $\eta_i(t) \geq \eta_{i,T}$  is automatic. If  $f_{i,T} = 1$ , scalar comparison with the constant solution  $f_i = 1$  implies that full automation is preserved, so  $\eta_i(t) = 1$ .

Define the fixed positive coefficient

$$d_i \equiv \begin{cases} p_i \eta_{i,T}^{\frac{\theta}{1-\theta}} \underline{L}_{AI,i}^{1-\eta_{i,T}}, & \eta_{i,T} > 0, \\ p_i \min\{L_{i,T}, \underline{L}_{AI,i}\}, & \eta_{i,T} = 0. \end{cases}$$

The second case follows from the initial CES lower bound and  $L_i(t) \geq L_{i,T}$ . Thus, in every sector,

$$\hat{L}_i(t) \geq d_i (C_i(t) S(t))^{\eta_{i,T}}.$$

*Applying the comparison argument.* Consider an economy starting from the same stocks at time  $T$ , with the same investment rates, depreciation rates, research parameters, and sectoral compute shares, but with effective labor given by

$$\hat{L}_i^{\text{low}}(t) = d_i (C_i^{\text{low}}(t) S^{\text{low}}(t))^{\eta_{i,T}},$$

where the superscript *low* denotes variables in this comparison economy. This is the Cobb-Douglas model of Corollary 3, where automation shares have been set to the elasticity at time  $T$ ,  $f_i = \eta_{i,T}$ ; with human labor held fixed at its date- $T$  level; and with fixed positive productivity coefficients.

The effective labor bounds above imply that, evaluated at the stocks of the CES economy, each stock's rate of accumulation is weakly higher under the CES model than under the comparison model. Further, in the comparison model, increasing any stock weakly increases the rate of accumulation of every other stock. Thus, starting from the same stocks at  $T$ , each stock in the CES economy remains at least as large as its counterpart in the comparison economy:

$$(A(t), S(t), H(t), C(t), K(t)) \geq (A^{\text{low}}(t), S^{\text{low}}(t), H^{\text{low}}(t), C^{\text{low}}(t), K^{\text{low}}(t))$$

for as long as both solutions exist.

Finally, (66) states that

$$\eta_{Y,T} + \eta_{S,T} r_S + \eta_{H,T} r_H + \frac{\eta_{A,T} r_A}{1 - \alpha} > 1.$$

By Corollary 3, the comparison economy therefore explodes in finite time. Since the CES economy dominates this comparison economy, it must also explode in finite time, no later than the comparison economy.  $\square$